Chemical Bonding

Chemical bond:

A chemical bond is a force of attraction between two or more species (i.e. atoms, ions, molecules) which hold them together and make them function as a single unit.

on the basis of dissociation energy, bonds are classified into two categories:

Strong Bond	Weak force
Energy required to break interaction is usually greater than or equal to 50 kJ/mol.	Energy required to break interaction is usually lower than 50 kJ/mol.
ex: Ionic bond Covalent bond Metallic bond	ex: Hydrogen bond van der waal's forces

≪ N≡N (Dinitrogen): Bond dissociation energy ≃ 945 kJ/mol (very high (∵ N_2 is very less reactive))

≪ CO (carbon monoxide): Bond dissociation energy ≃ 1046 kJ/mol

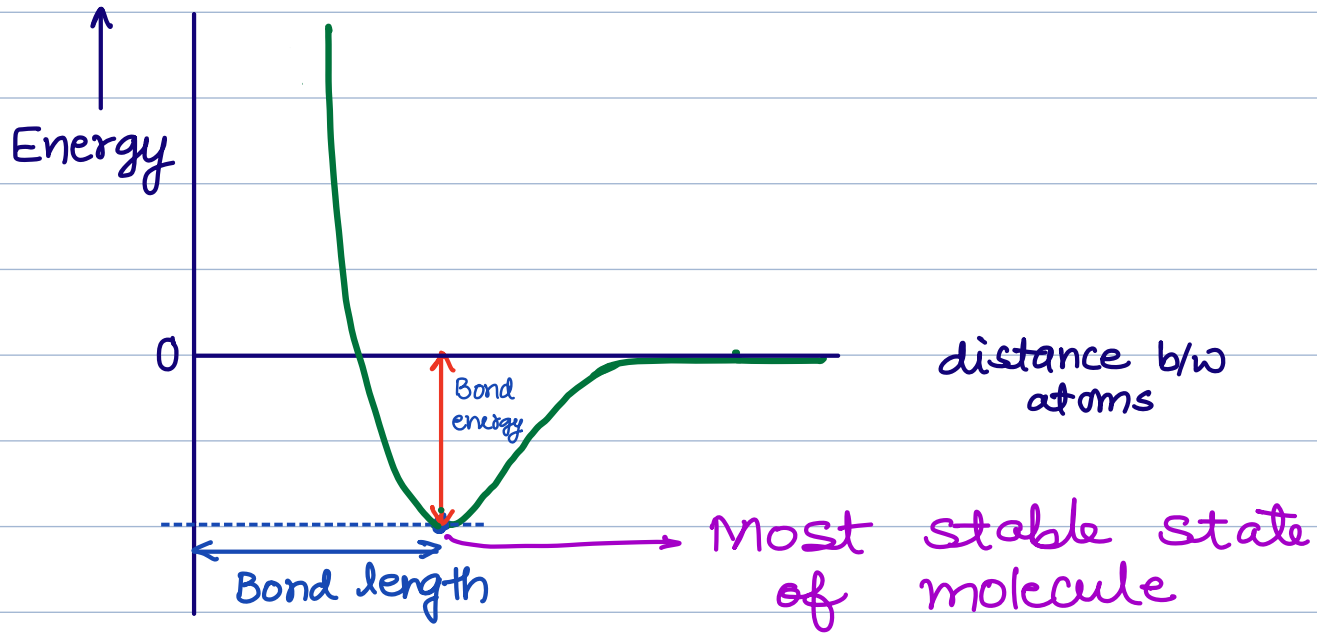
↳ [Max. bond dissociation energy among diatomic molecules]

[Noble gas elements have 8 e^- in their outermost shell. Noble gases are stable and almost non-reactive, so it was thought that other elements try to attain 8 e^- in their outermost shell to become stable like noble gases. But it was failed for many species]

Energy change in bond formation process:

$$\underset{\text{(element)}}{A} + \underset{\text{(element)}}{A} \longrightarrow \underset{\text{(molecule)}}{A_2}$$

- Energy of molecule A_2 will be lower as compared to that of A.
- ΔH for bond formation process = -ve (Bond formation is always an exothermic process).



How to draw Lewis structures:

(1). Selection of central atom:

usually the element which is less in number in the given formula is central atom but if more than 1 atoms are these which are least numbered then least electronegative atom is chosen as central atom.

[H and F can't be central atom as their covalencies are 1]

ex: $\overset{*}{C}H_2FCl$ (central atom) ; $\overset{*}{N}H_3$ (central atom) ; $\overset{*}{C}_2H_4Cl_2$ (central atom)

(2). Arrange surrounding atoms around central atom and connect them with central atom through no. of bonds as per valencies of central and surrounding atoms.

(3). Show lone pairs (if available) on central atom and surrounding atoms.

Lone pair (LP): valence e^- pairs which are not used in bond formation

Bond pair (BP): electron pairs of covalent bond.

(4). Try to avoid 3-membered cycle (ring) and peroxide linkage (O–O). (O–O: single bond)

(5). central atom can be more than one in number.

*(6). covalency of surrounding halogen is 1.

(7). Try to put +ve charge on less electronegative element and -ve charge on more electronegative element.

(8). In case of species having more than one central atom, try to make a symmetrical structure. (not correct always).

(9). If covalency of central atom or surrounding atom is satisfied, they do not possesses charge.

(10).

species in which:

No. of central atoms ≥ 3

No. of surrounding atoms $\geq$ No. of central atoms

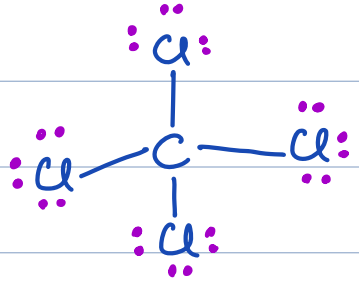
There is very high probability of cyclic structure.

(11). Try to avoid making double or triple bond between atoms having period no. more than or equal to 3.

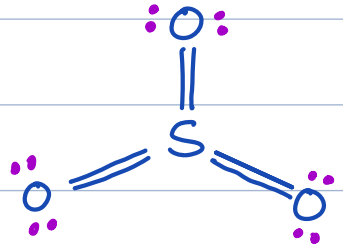
(12). Try to put as less charge on an atom as you can in given molecule.

examples of non-charged species:

(i). CCl_4

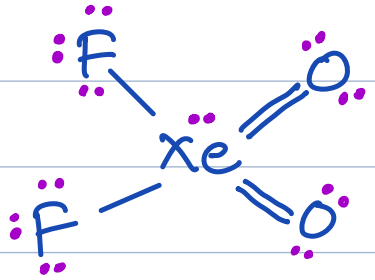


(ii) SO_3

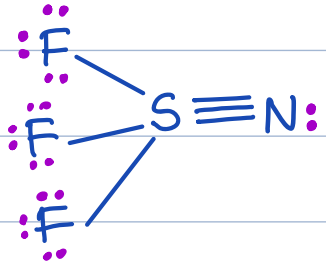


(iii). XeF_2

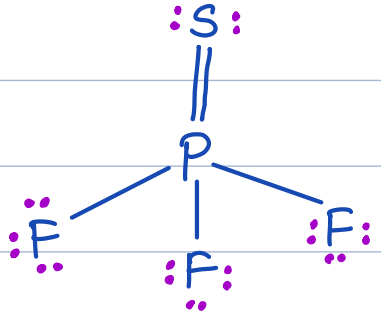
(iv). XeO_2F_2



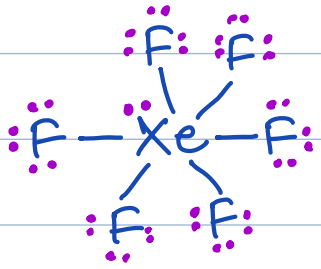
(v). SNF_3



(vi). PSF_3

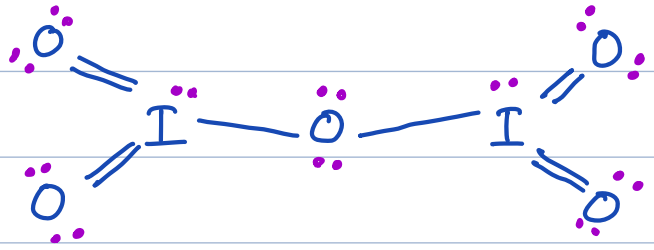


(vii). XeF_6

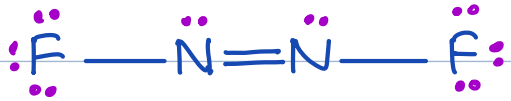


(viii). C_2H_2

(ix). I_2O_5

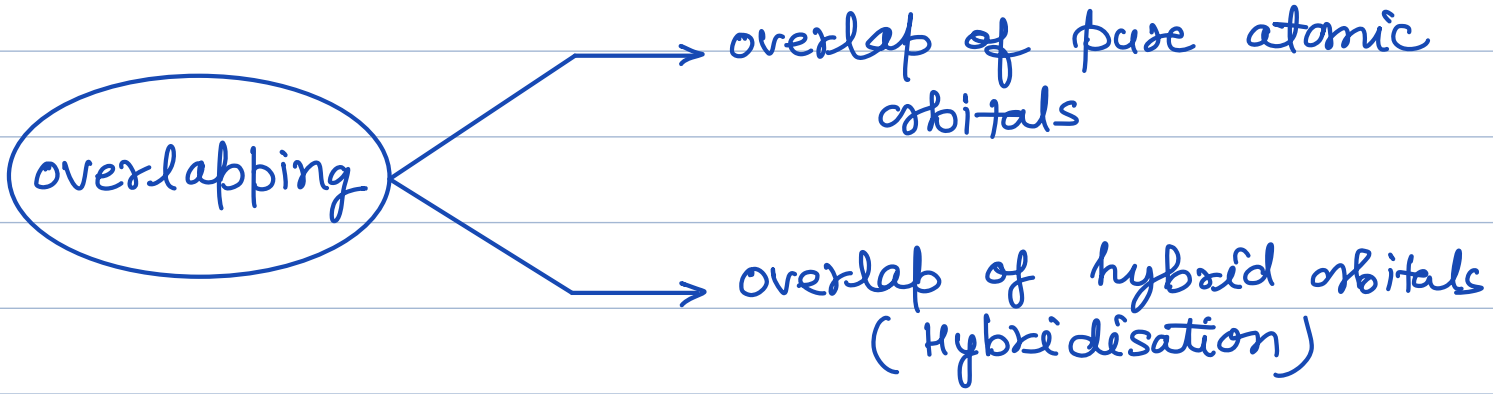


(x). N_2F_2



Valence Bond Theory (VBT):

Overlapping : Increase in electron probability density between the nuclei of combining atoms is called overlapping.

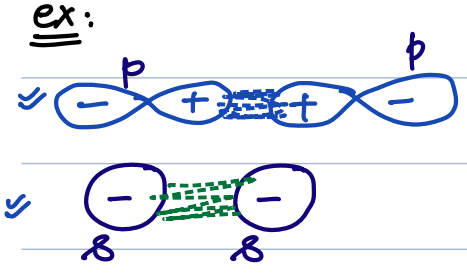
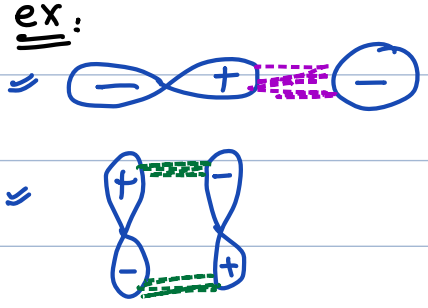
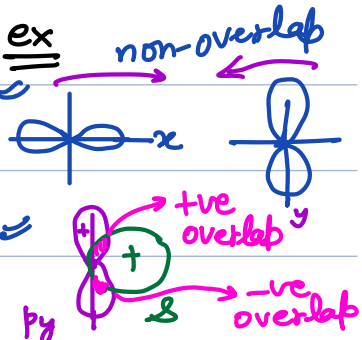


overlapping type:

(i) Head on ; Head to head ; Axial overlap

(ii) Lateral overlap, sideways overlap

Results in → bond formation

Positive overlap	Negative overlap	Zero overlap
overlap of lobes of same sign.	overlap of lobes of opposite sign	non-overlap (or) net zero overlap
ex: 	ex: 	ex 

≈ The bond formed by head on overlap of atomic orbitals is called sigma bond (σ).

≈ The bond formed by sideways overlap of atomic orbitals is called Pi- bond. (π).

Overlap of atomic orbitals:

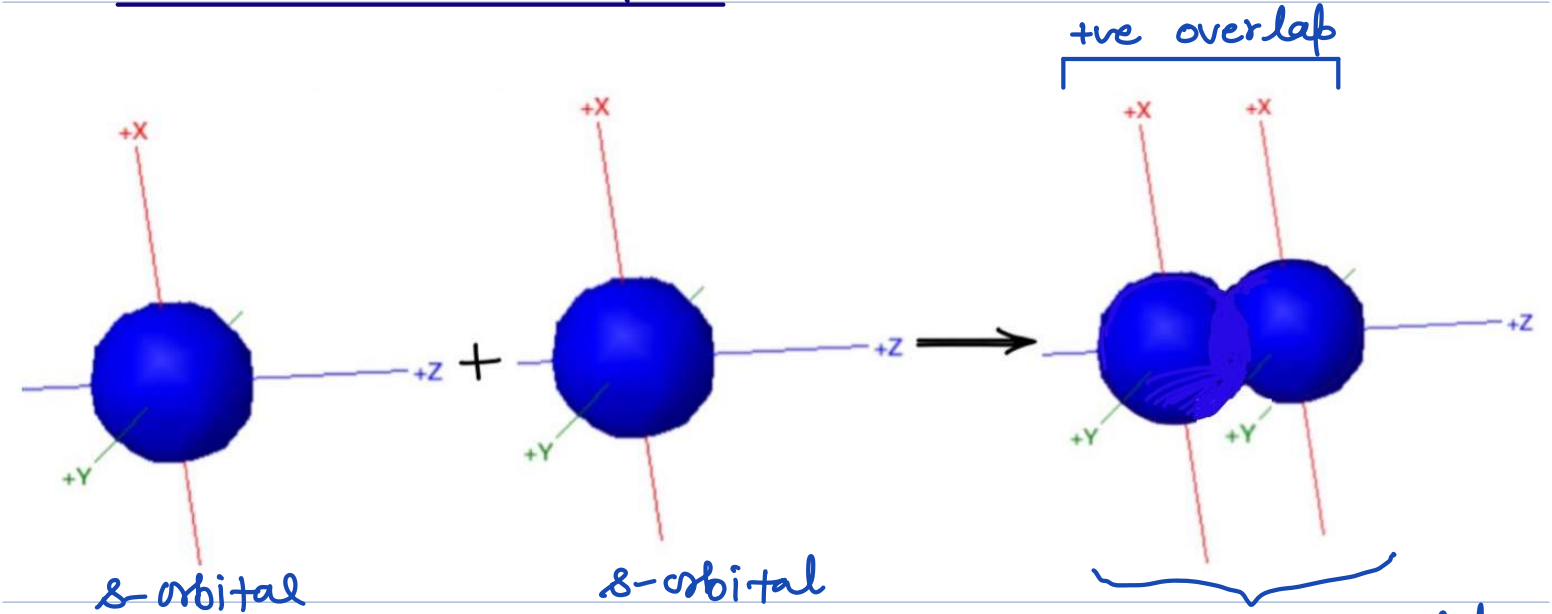
- **Internuclear axis (INA) =** It is the line along which two orbitals approach towards each other for overlap.

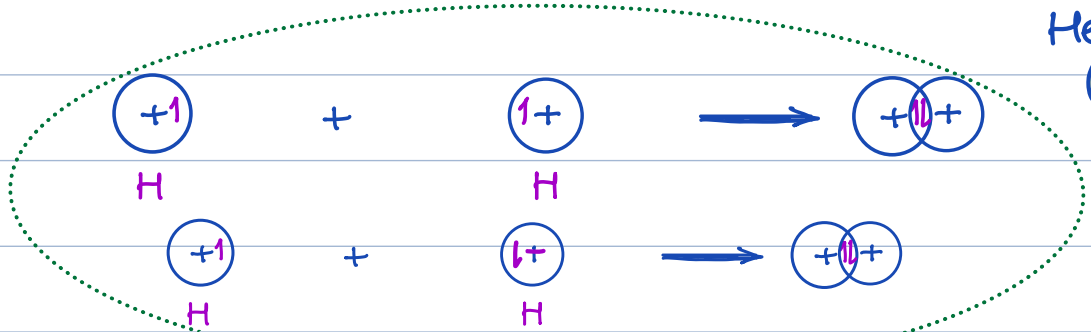
- Conventionally, Z-axis is taken as internuclear axis.

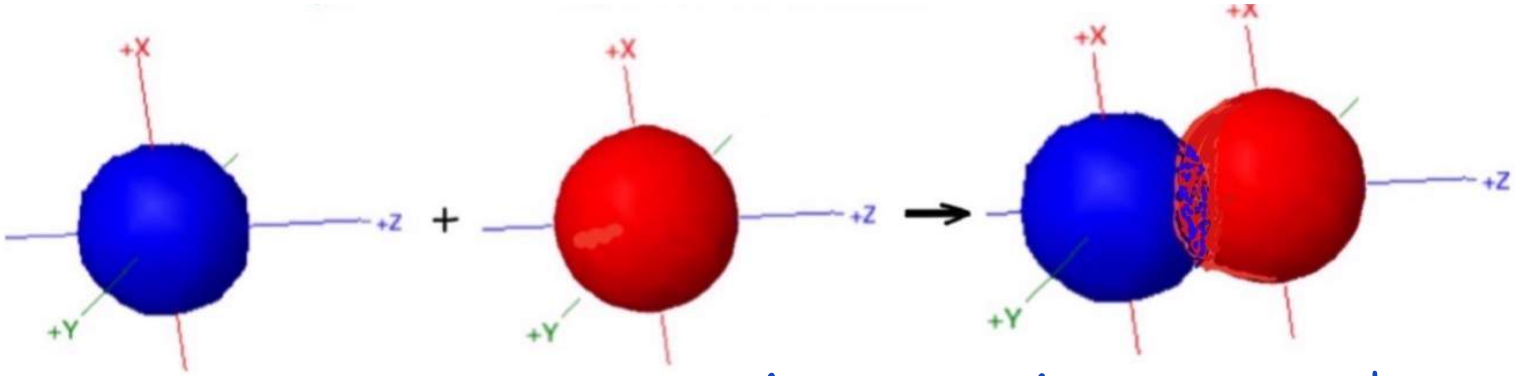
Various types of overlaps:

} +ve } -ve

(i) s–s overlap:



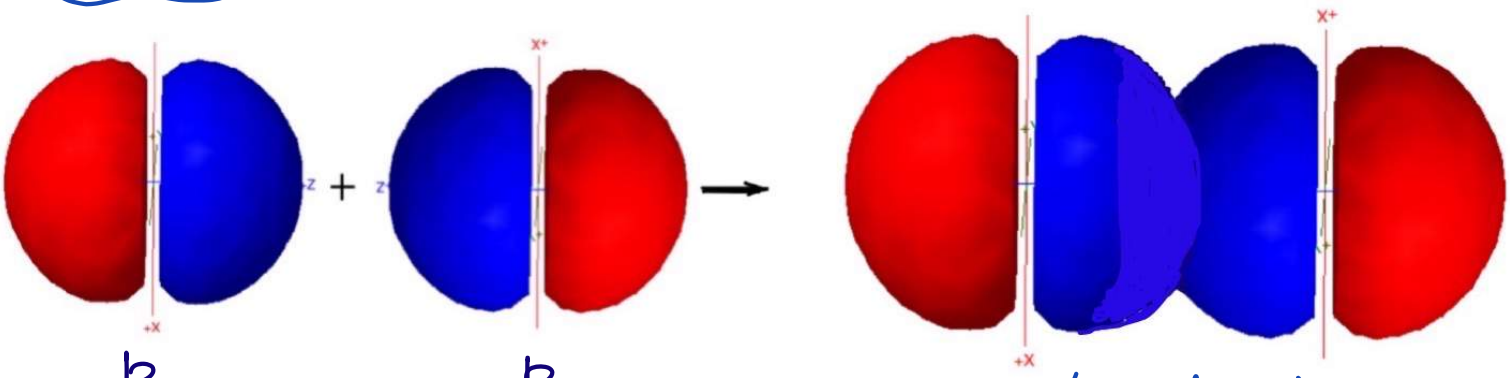




s-orbital s-orbital (Head on negative overlap)

(ii) p – p overlap:

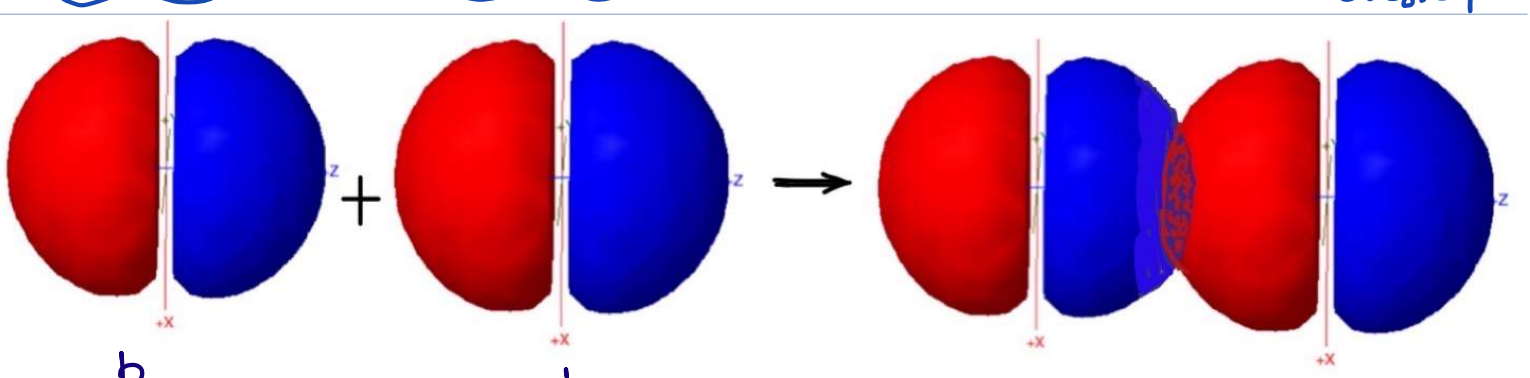
(+ive overlap)



p_z p_z (Head on)

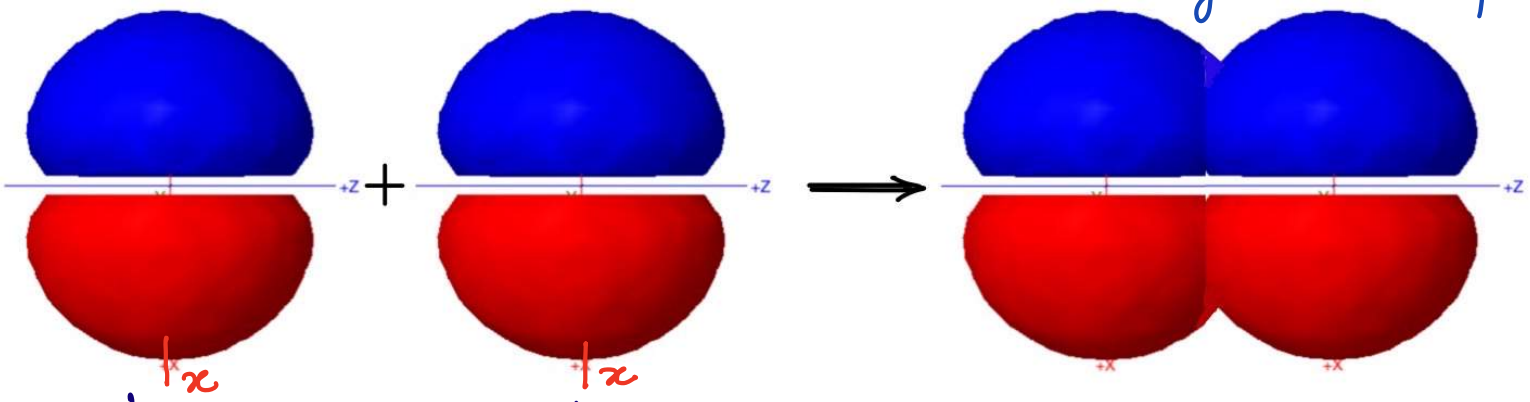
(σ – bond)

-ve overlap

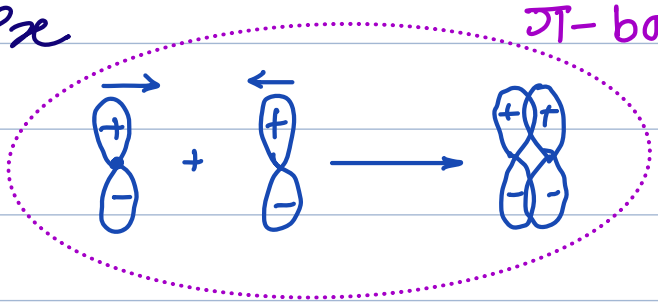


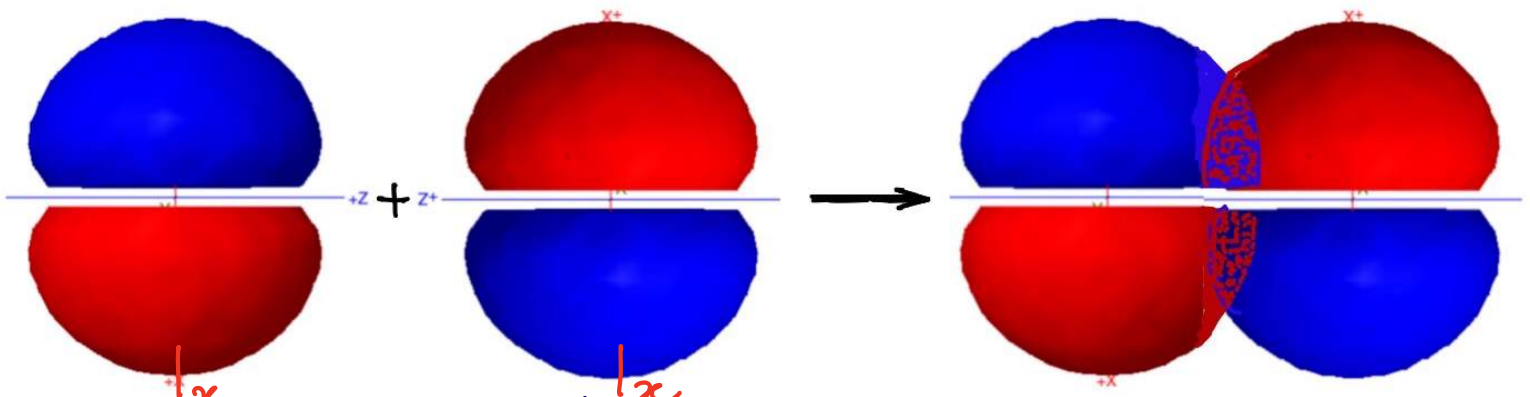
p_z p_z (Head on)

sideways +ve overlap

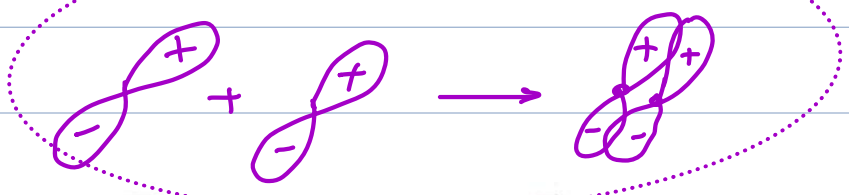


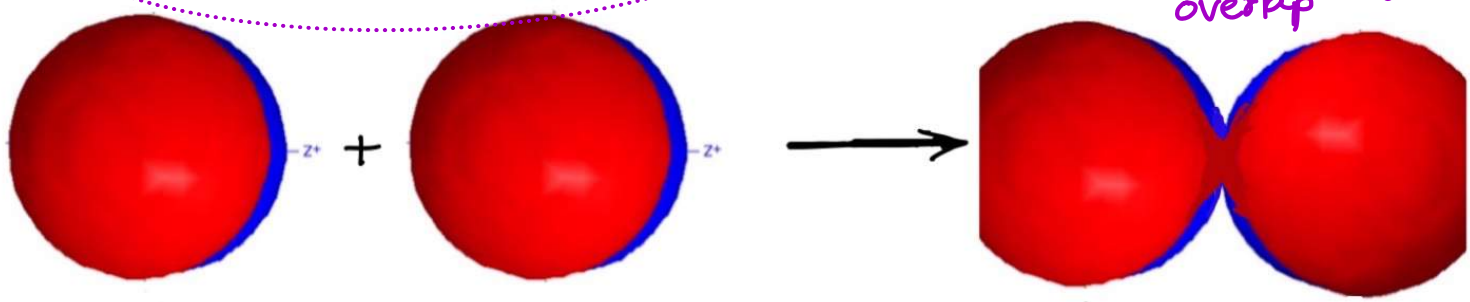
p_x p_x π- bond





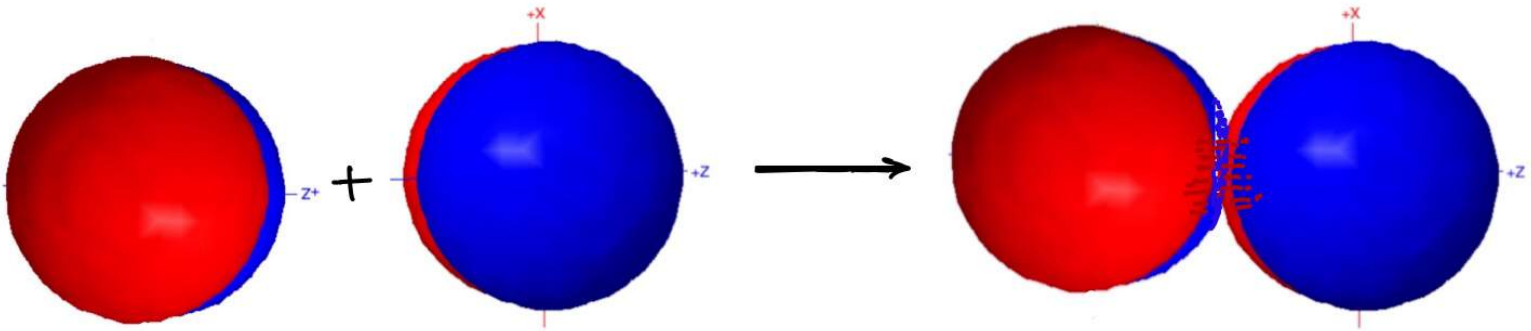
p_x p_x (sideways -ve overlap)





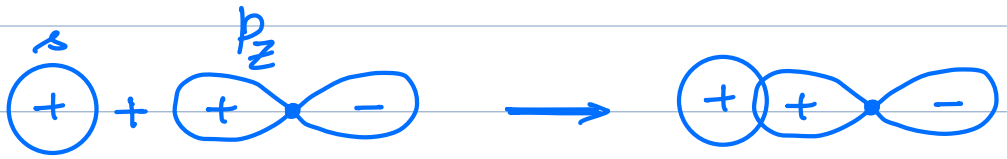
p_y p_y (π- bond)

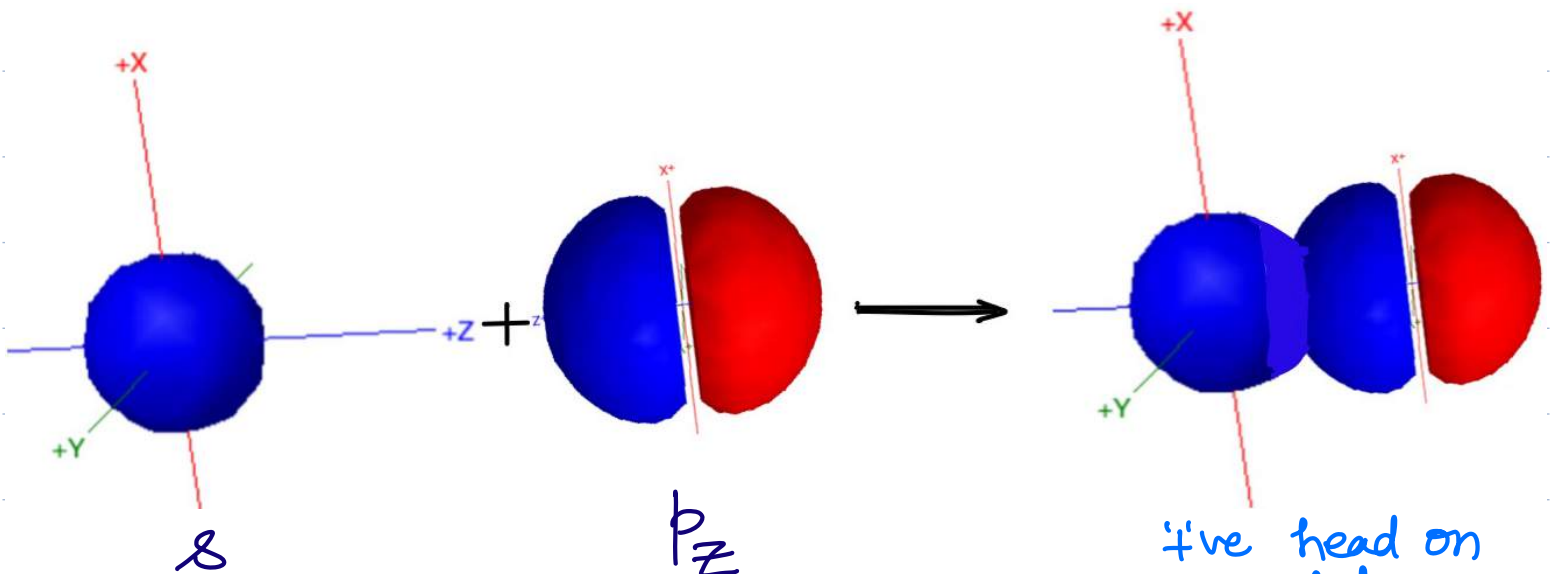
Positive sideways overlap



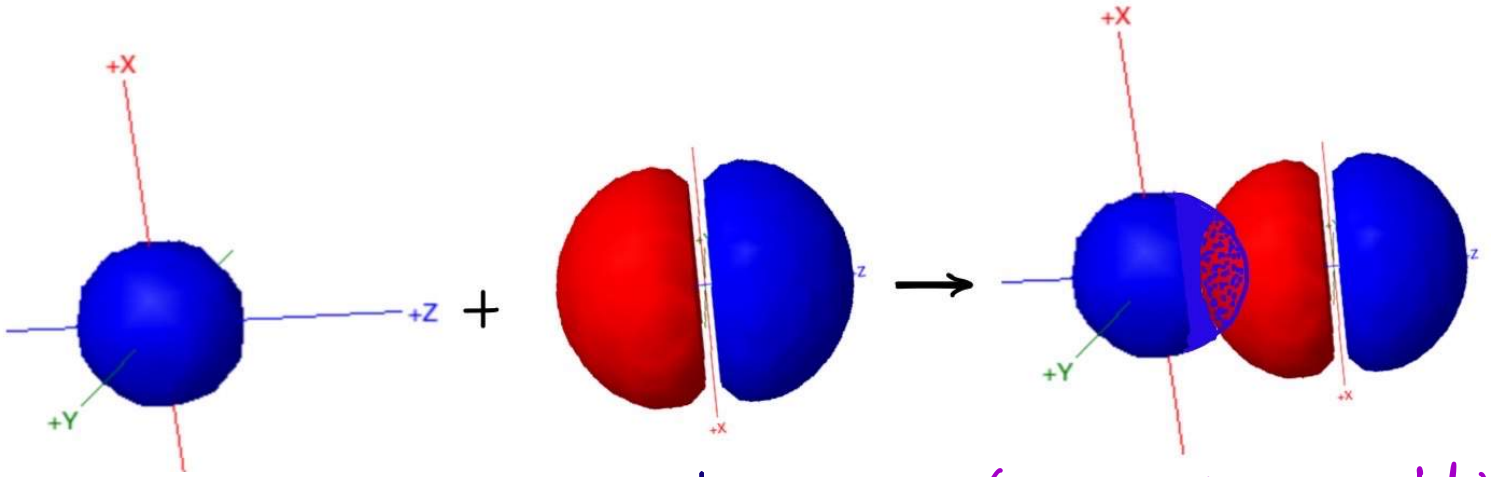
p_y p_y (-ve sideways overlap)

(iii) s–p overlap:

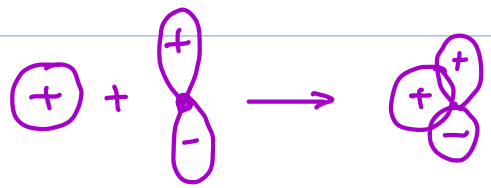


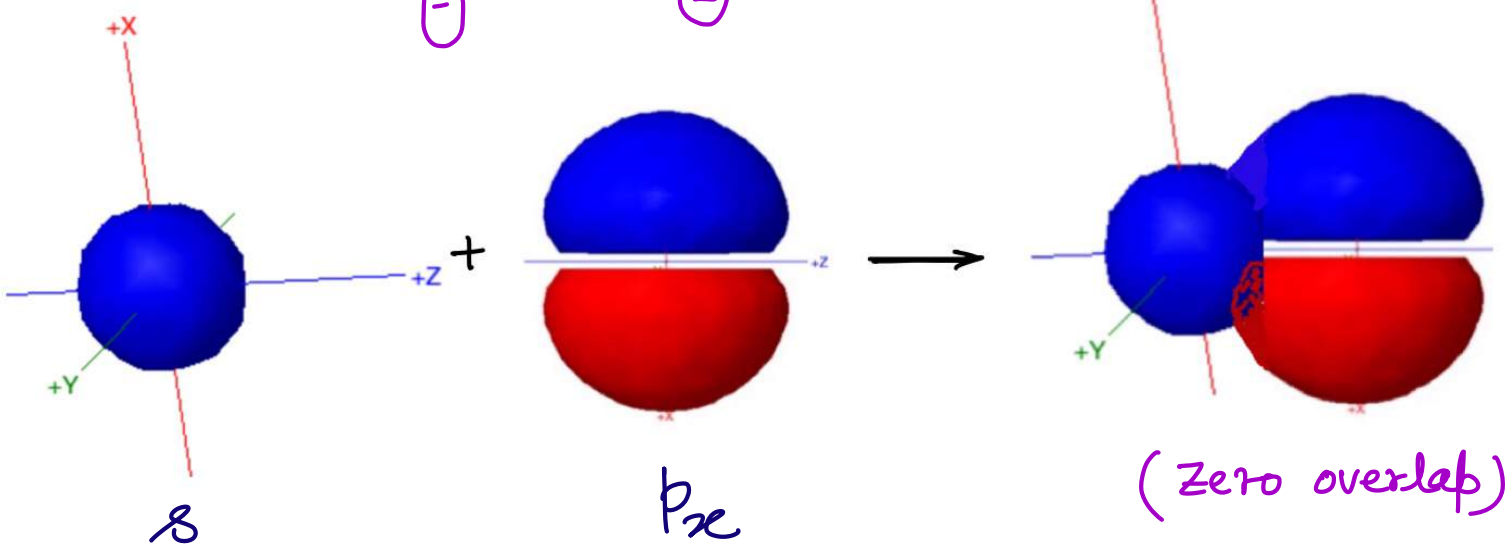


s p_z +'ve head on overlap (σ-bond)

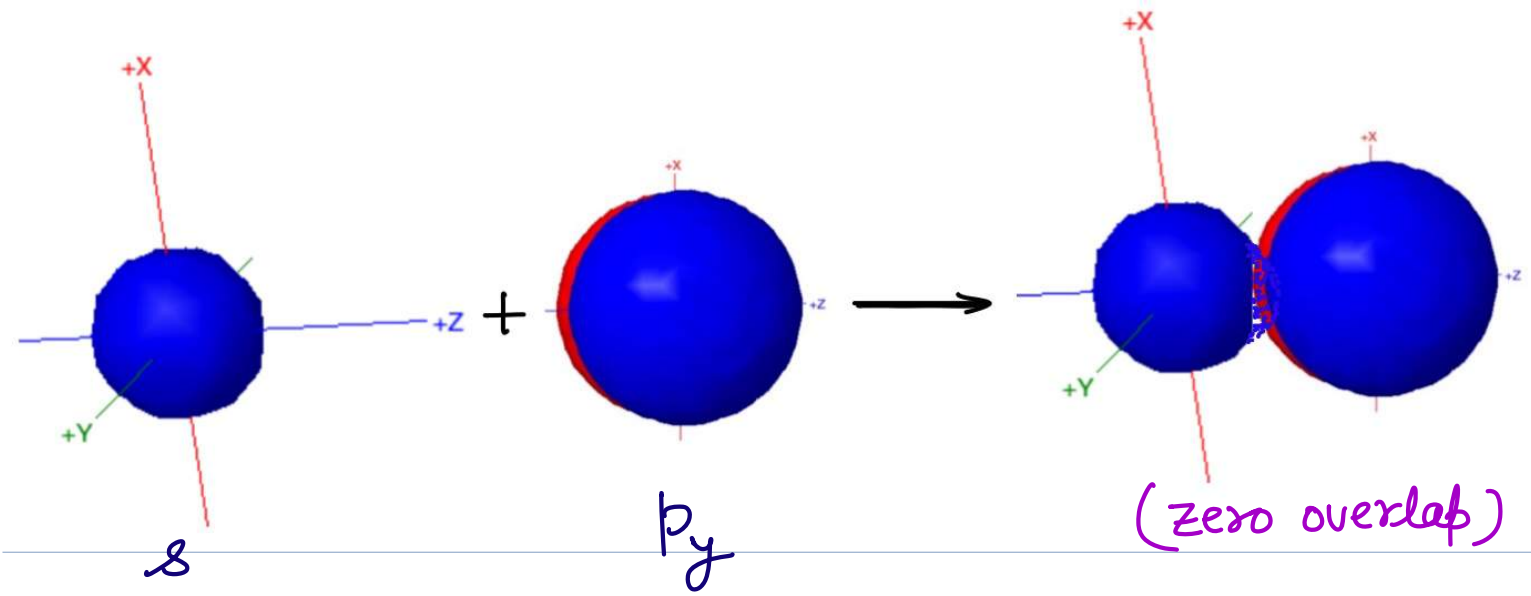


s p_z (-ve Head on overlap)





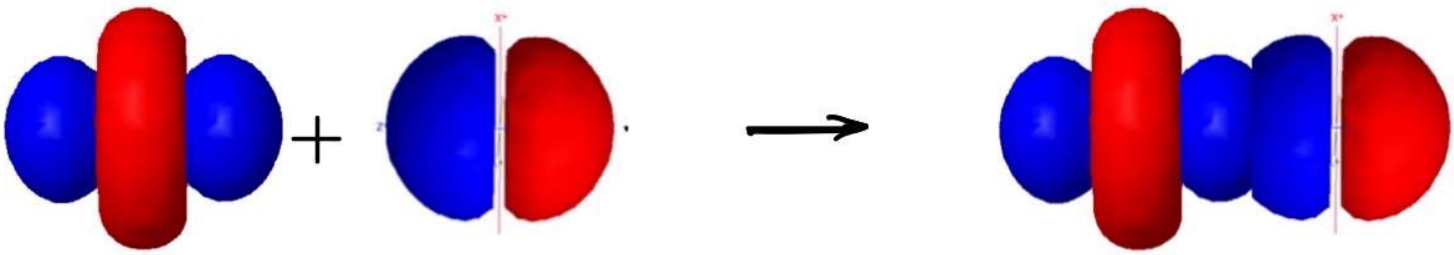
s p_x (Zero overlap)



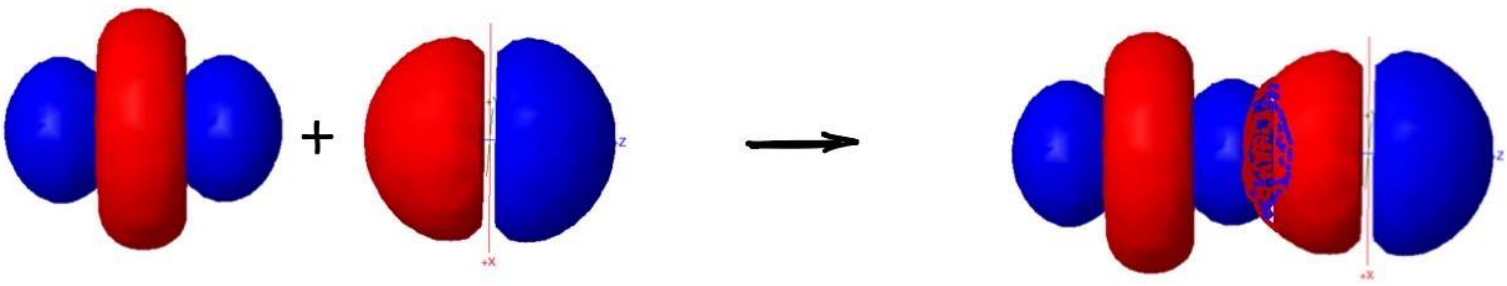
s p_y (Zero overlap)

(iv) p–d overlap :

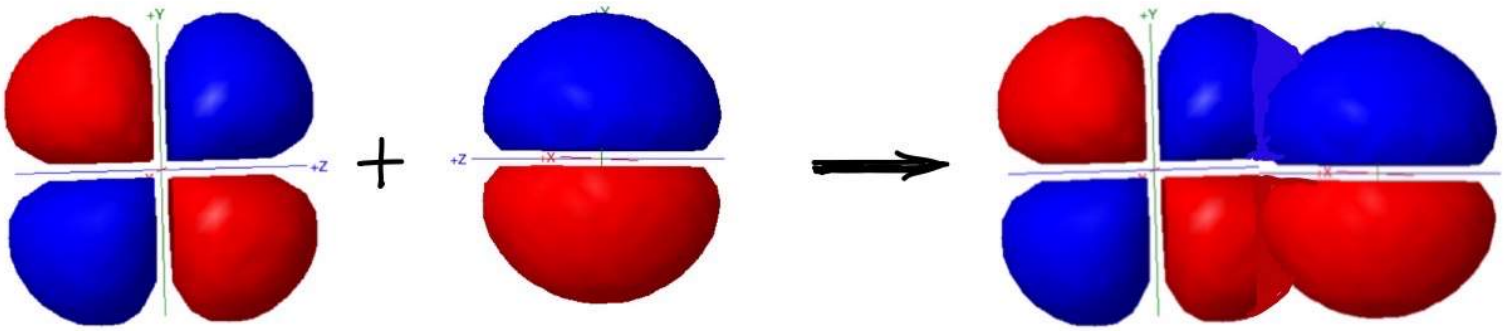
+ve head on overlap



d_{z^2} p_z (σ– bond)

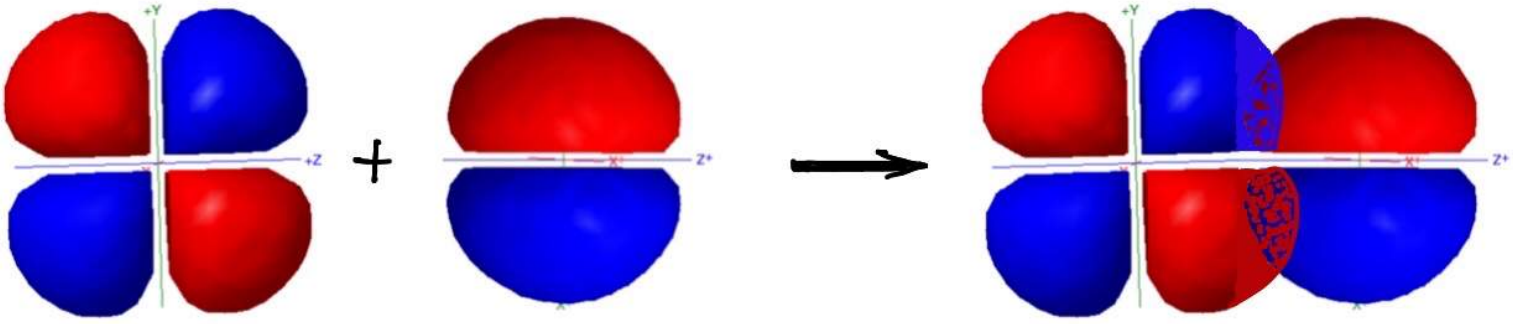


d_{z^2} p_z (-ve head on overlap)

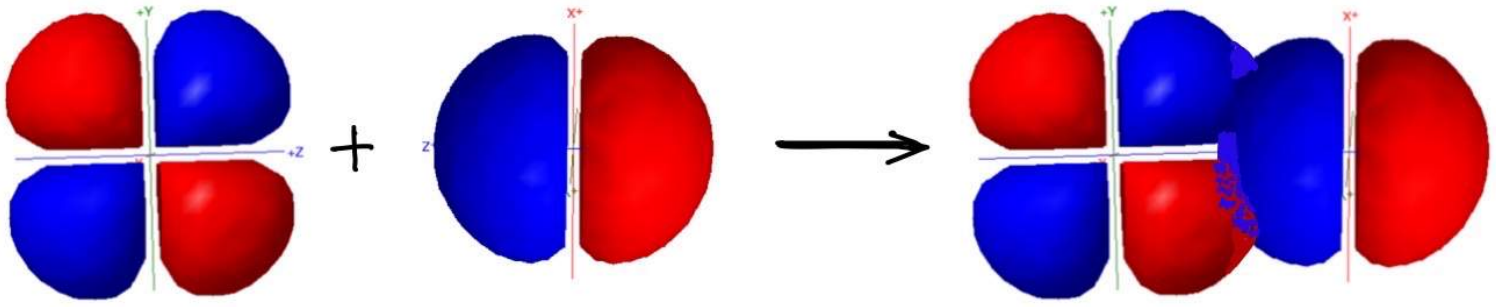


d_{yz} p_y (+ve sideways overlap)

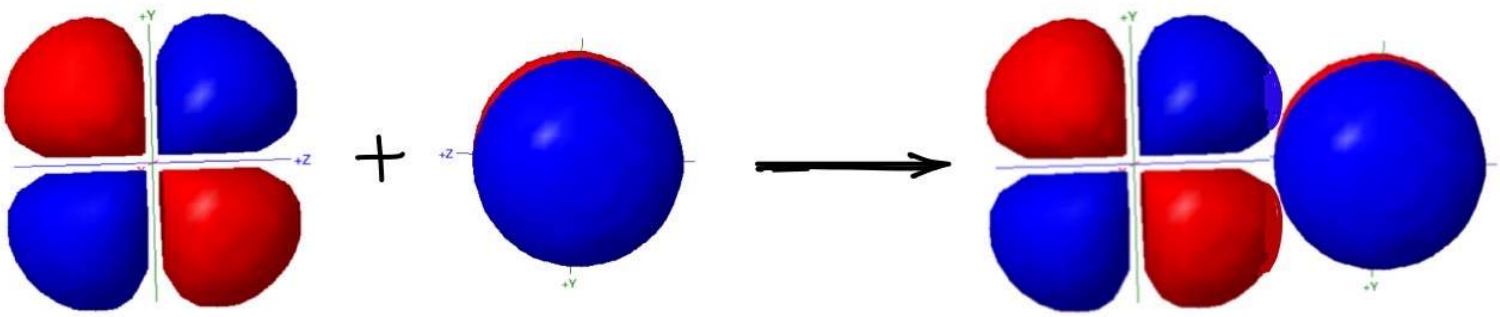
(π– bond)



d_{yz} p_y (-ve sideways overlap)

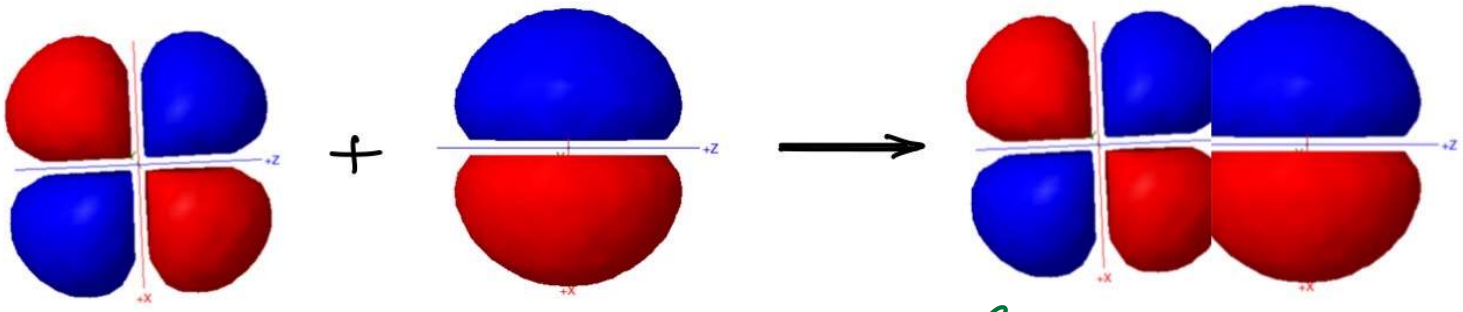


d_{yz} p_z (zero overlap)



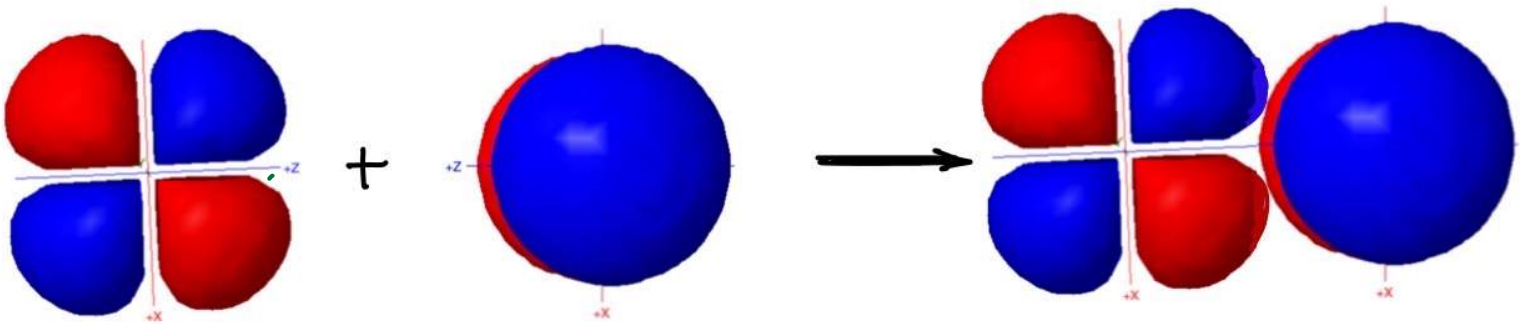
d_{yz} p_x (zero overlap)

(π- bond)



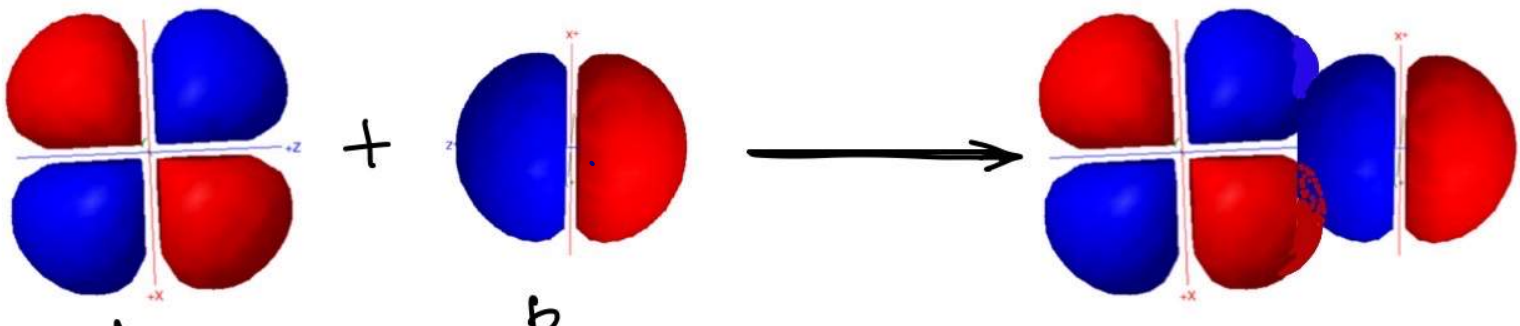
d_{xz} p_z

(+ve sideways overlap)



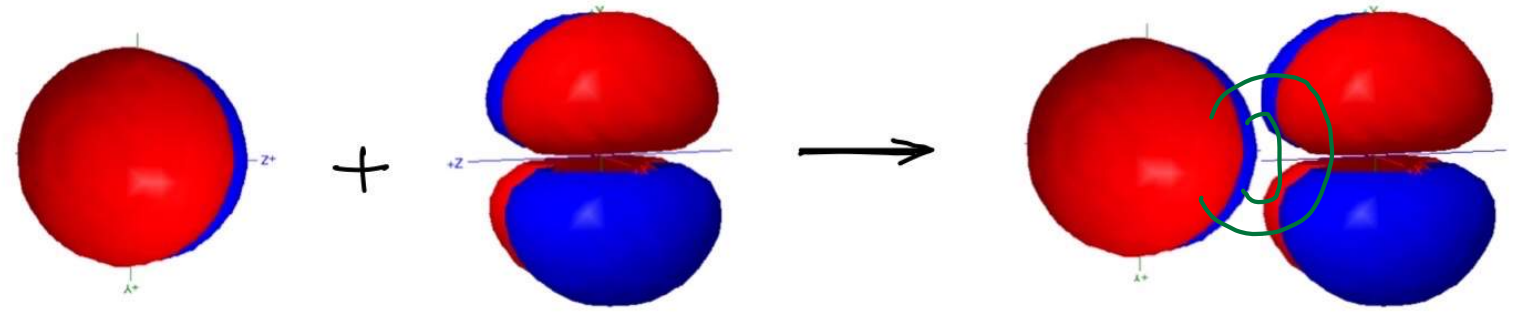
d_{xz} p_y

(zero overlap)

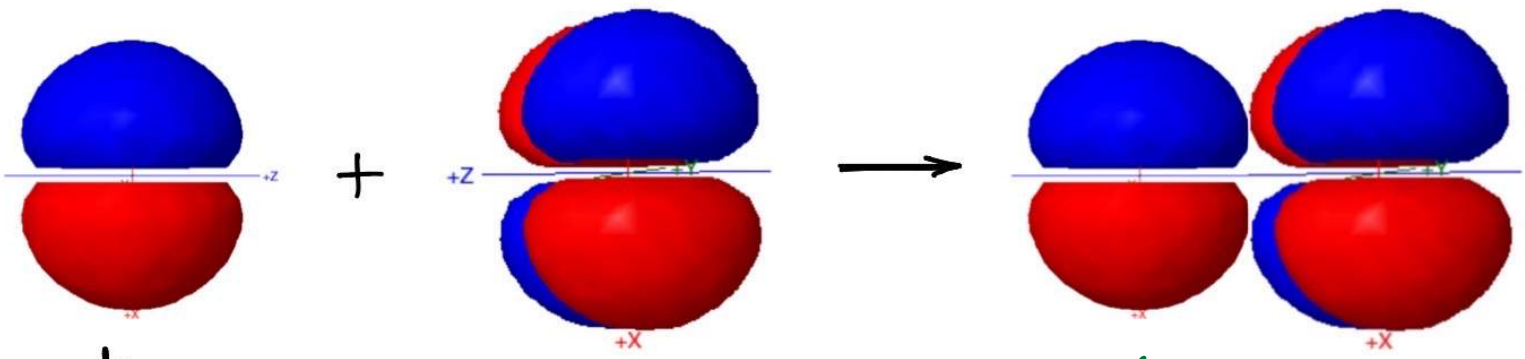


d_{xz} p_z

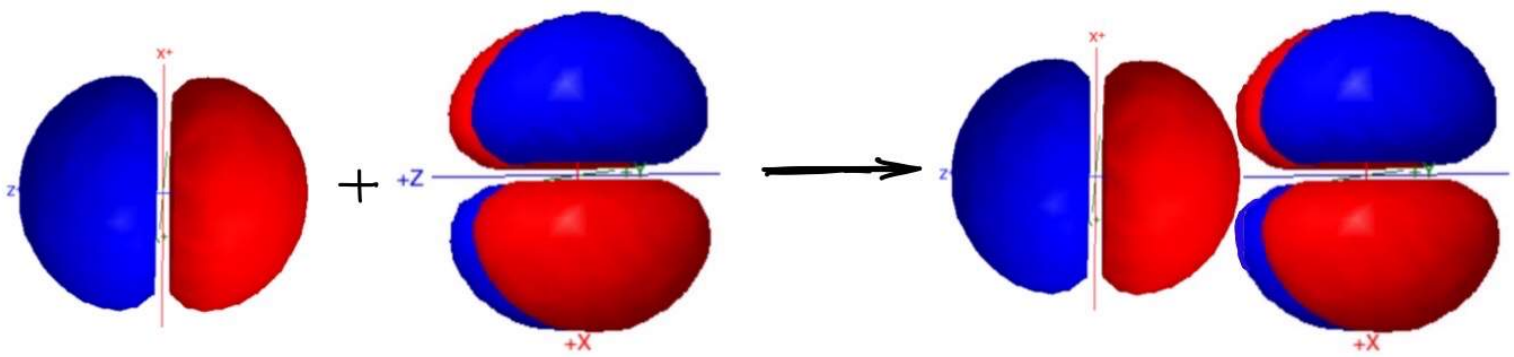
(zero overlap)



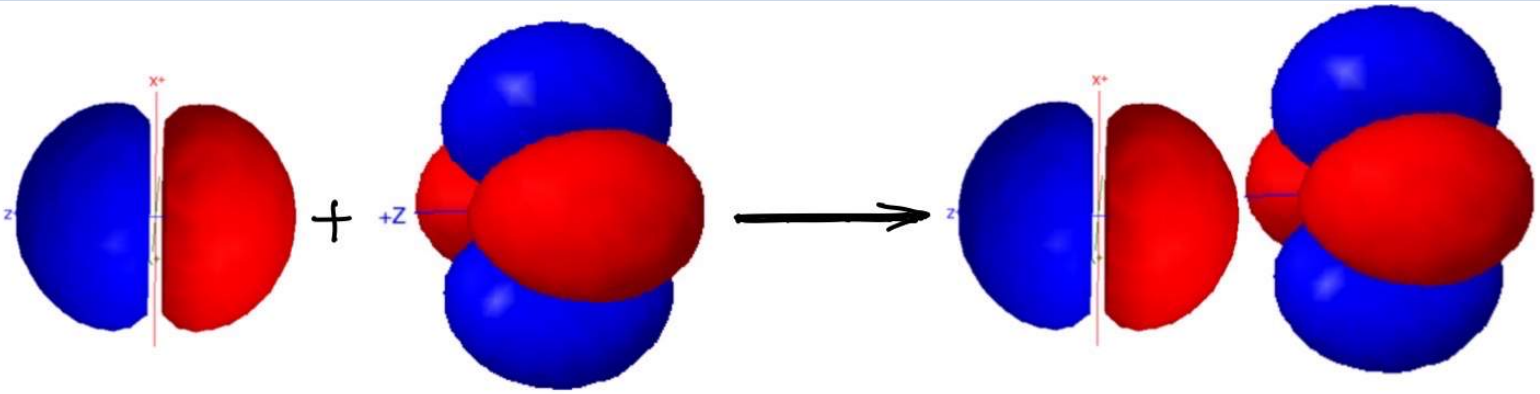
p_y d_{xy} (zero overlap)



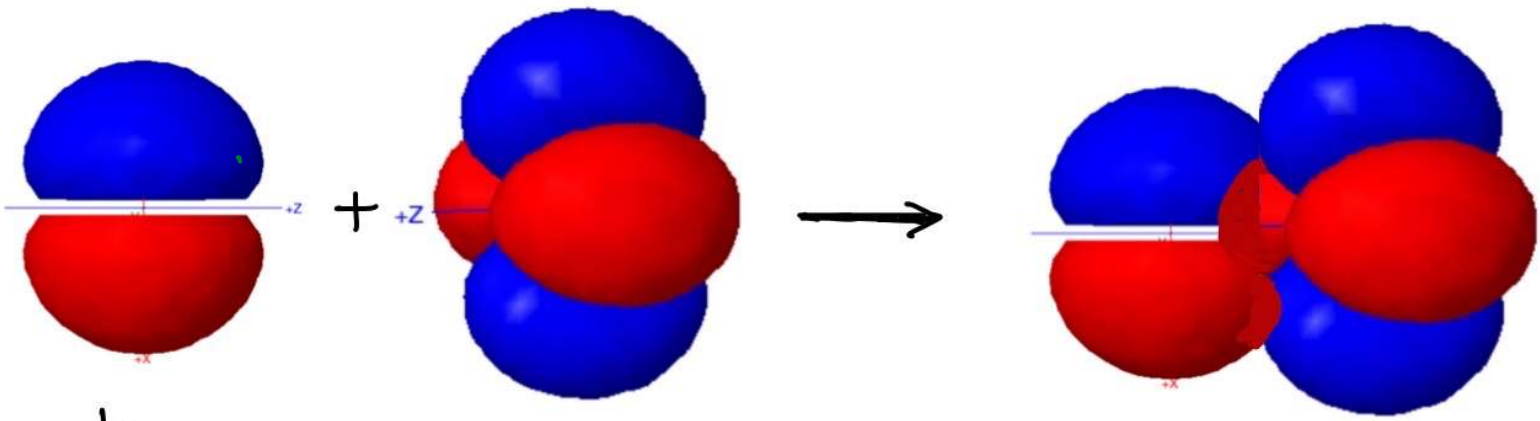
p_z d_{xy} (zero overlap)



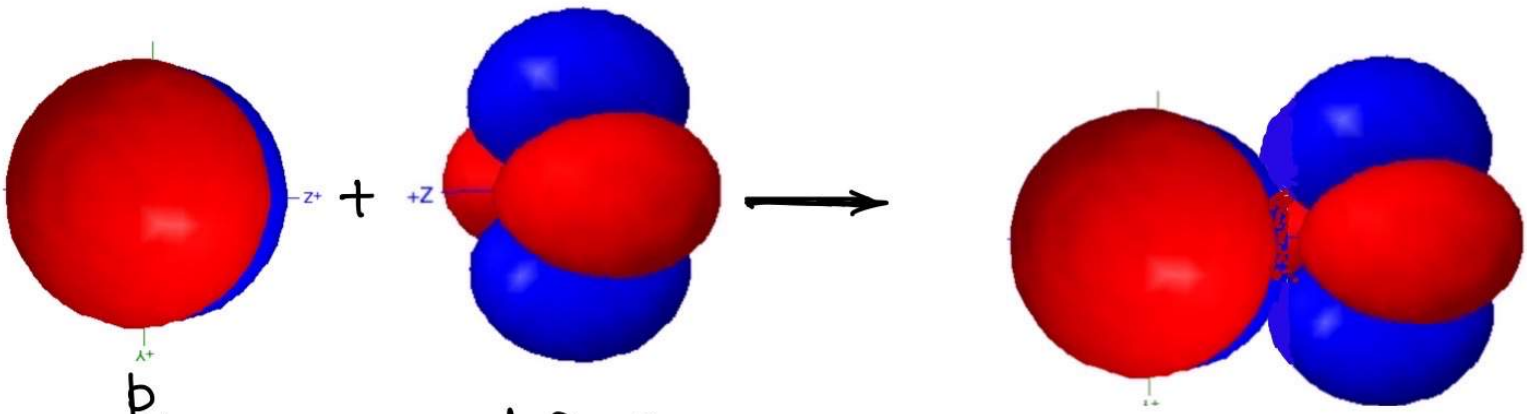
p_z d_{xy} (zero overlap)



p_z $d_{x^2-y^2}$ (zero overlap)



p_x $d_{x^2-y^2}$ (zero overlap)



p_y $d_{x^2-y^2}$ (zero overlap)

Types of covalent bonds :

(i) **σ – bond** The overlap of one lobe with one lobe result into σ – bond.

(ii) **π – bond** The overlap of two lobes with two lobes result into π – bond.

(iii) **δ – bond** The overlap of four lobes with four lobes result into δ – bond.

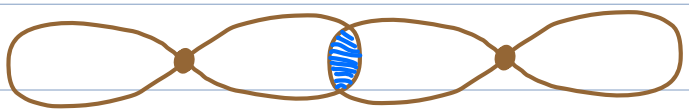
(iv) **ϕ – bond :** The overlap of six lobes with six lobes result into ϕ – bond.
(not in our syllabus)

Bond strength comparisons:

(i). More is the area of overlapping between orbitals, stronger will be the bond formed.

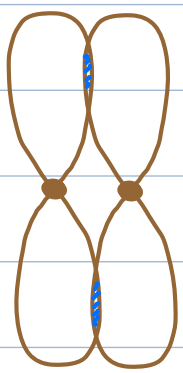
ex: v.imp. A sigma bond is usually stronger than a π-bond.

(σ)



more area of overlap

(π)



Less area of overlap

(ii). With increasing size of orbitals, electron probability density decreases and hence bond strength decreases.

ex:

- $\underset{\text{inc. bond strength}}{\overset{(\sigma)}{1s-1s} > \overset{(\sigma)}{1s-2s} > \overset{(\sigma)}{2s-2s} > \overset{(\sigma)}{3s-3s}}$

dec. internuclear distance

- $\underset{\text{inc. bond strength}}{2p\pi-2p\pi > 2p\pi-3p\pi > 3p\pi-3p\pi}$

(iii). np-orbital forms stronger sigma bond as compared to ns-orbital. (for same n-value).

ex:

$np-np$ (σ) > $ns-np$ (σ) > $ns-ns$ (σ)

inc. bond strength

Reason:

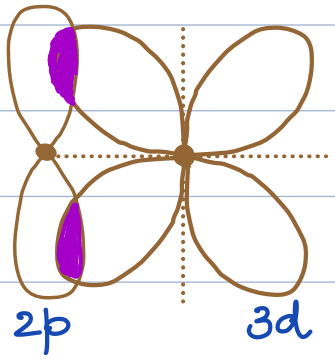
p-orbital is more directional in nature as compared to the s-orbital.

(iv).

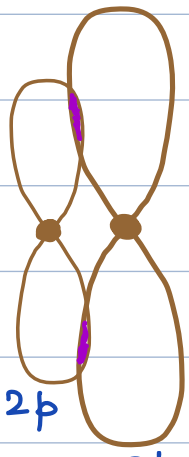
inc. bond strength

$2p\pi-2p\pi$ > $2p\pi-3d\pi$ > $2p\pi-3p\pi$

explanation



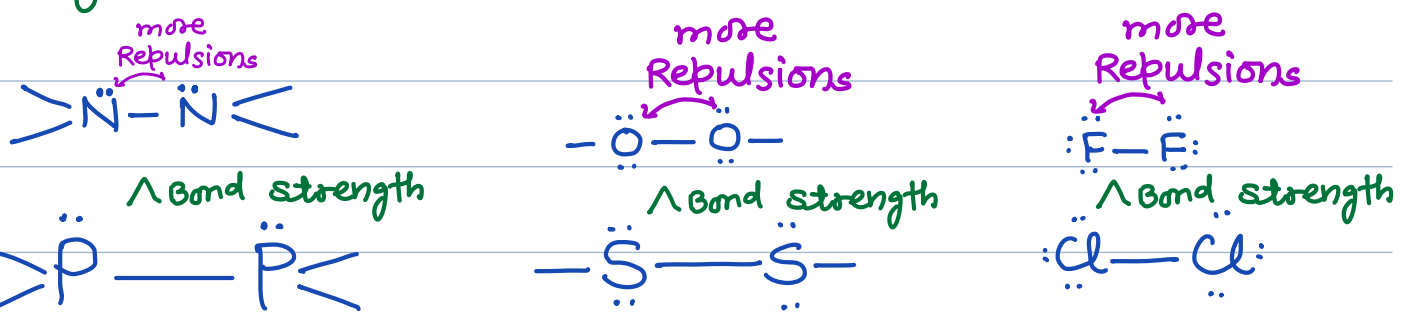
"More area of overlap due to tilted lobes of 3d"



"Less area of overlap due to parallel lobes of 3p"

(v) Due to repulsion of lone pairs of bonded atoms, the strength of single bond decreases.

ex:



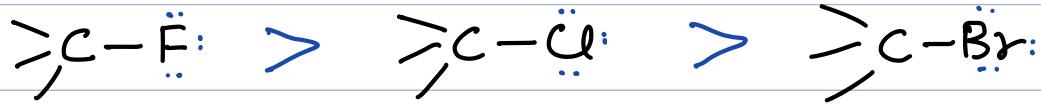
more repulsion b/w LP of second period ⟶ ∴ Shorter bond length

- In case of double or triple bonds, bond strength is Judged by internuclear distance

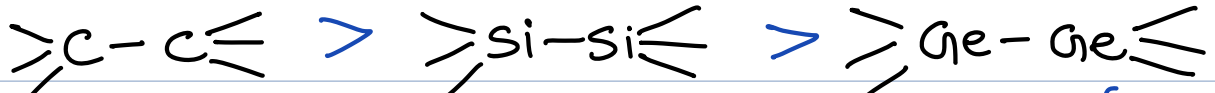
ex:

- ✓ $\ddot{O}=\ddot{O}$ ($2p\pi$-$2p\pi$) > $\ddot{S}=\ddot{S}$ ($3p\pi$-$3p\pi$) (Bond strength)
- ✓ $\ddot{N}\equiv\ddot{N}$ ($2p\pi$-$2p\pi$) > $\ddot{P}\equiv\ddot{P}$ ($3p\pi$-$3p\pi$) (Bond strength)
- ✓ $-\ddot{N}=\ddot{N}-$ > $-\ddot{P}=\ddot{P}-$ (Bond strength)

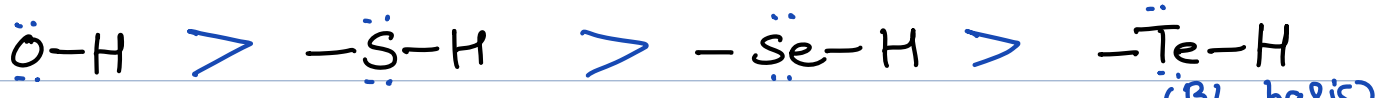
compare following bond strengths:

(i)  (BL basis)

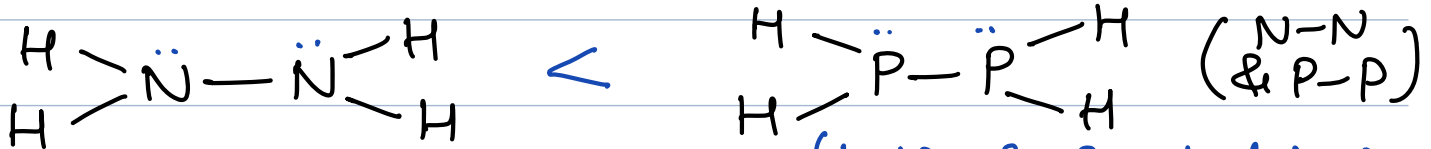
(ii)
(BL basis)

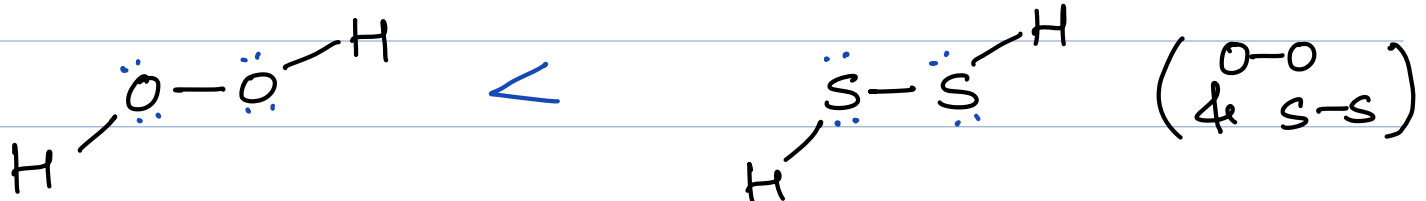
(iii) 
(BL basis)

(iv)
(BL basis)

(v) 
(BL basis)

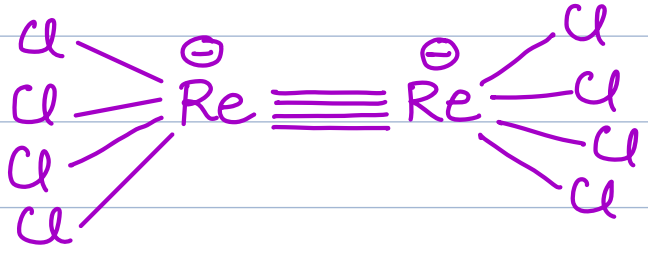
(vi)
(BL basis)

(vii) 
(less LP-LP repulsion)

(viii) 
(less LP-LP repulsion)

Generally,

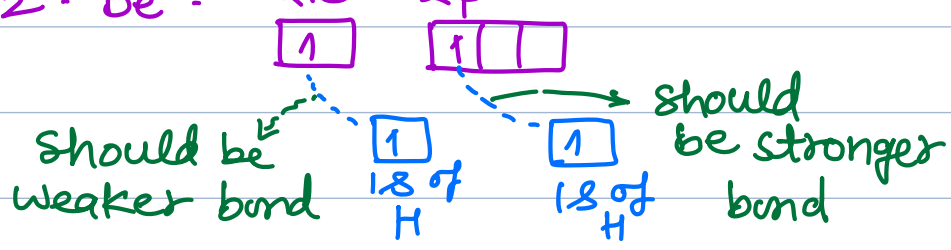
Bond	contains	example
single bond	one σ	H–H, Cl–Cl,
Double bond	one σ + one π	O=O, S=O
Triple bond	one σ + Two π	C≡C, N≡N
Quadruple bond	one σ + Two π + one δ	$[Re_2Cl_8]^{2-}$ ex. not to remember



Drawbacks of VBT (Pure orbital overlap):

- ✓ Unable to explain shape of many molecules.
- ✓ Unable to explain bond strengths in many.

ex: BeH_2 : Be : $2s^1$ $2p^1$



But in BeH_2, both bonds are of equal strength.

Concept of Hybridisation:

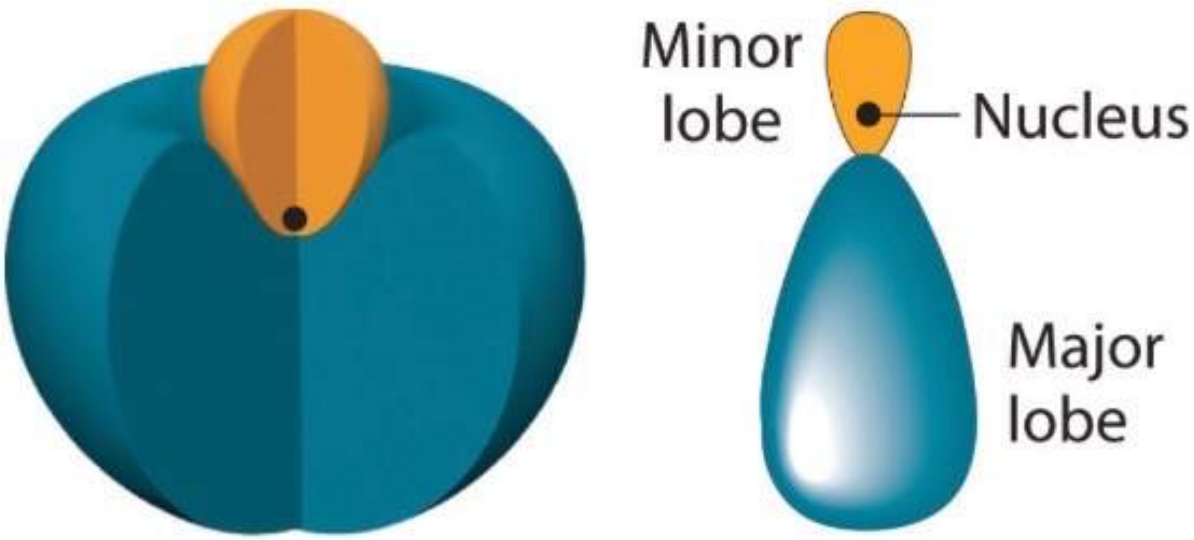
- It is mathematical manipulation of wave equation of pure atomic orbitals.

- The process of intermixing of two or more atomic orbitals of nearly same energy and re-distribution of energy among themselves to give equal no. of new orbitals (known as hybrid orbitals) having same shape, same size, same energy and different orientation in 3D-space is called hybridisation.

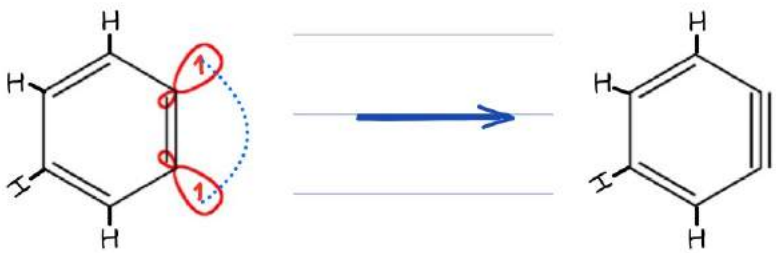
Some imp. points:

(i). Number of atomic orbitals involved in intermixing = Number of hybrid orbitals formed

(ii). Hybrid orbitals are unsymmetrical in shape. (one lobe is larger and other is shorter)



(iii). Hybrid orbitals form only sigma bond except in case of benzyne.



(iv). A hybrid orbital has more directional nature as compared to a pure atomic orbital.

(v). Hybridisation is intermixing of orbitals not of electrons.

(vi). vacant orbital, fully filled orbital, exactly half-filled orbital : any of these can involve in hybridisation.

(vii). In hybridisation, orbitals of same element are involved.

(viii). A hybrid orbital can overlap with atomic or hybrid orbital.

(ix). Number of hybrid orbitals on central atom:

= No. of σ-bonds + (No. of localised lone pair(s))

↳ Lone pair available on central

Mixing of orbitals	Name of hybrid orbital formed
one 's' + one 'p'	sp
one 's' + two 'p'	sp^2
one 's' + three 'p'	sp^3
one 's' + three 'p' + one 'd'	sp^3d
one 's' + three 'p' + two 'd'	sp^3d^2
one 's' + three 'p' + three 'd'	sp^3d^3

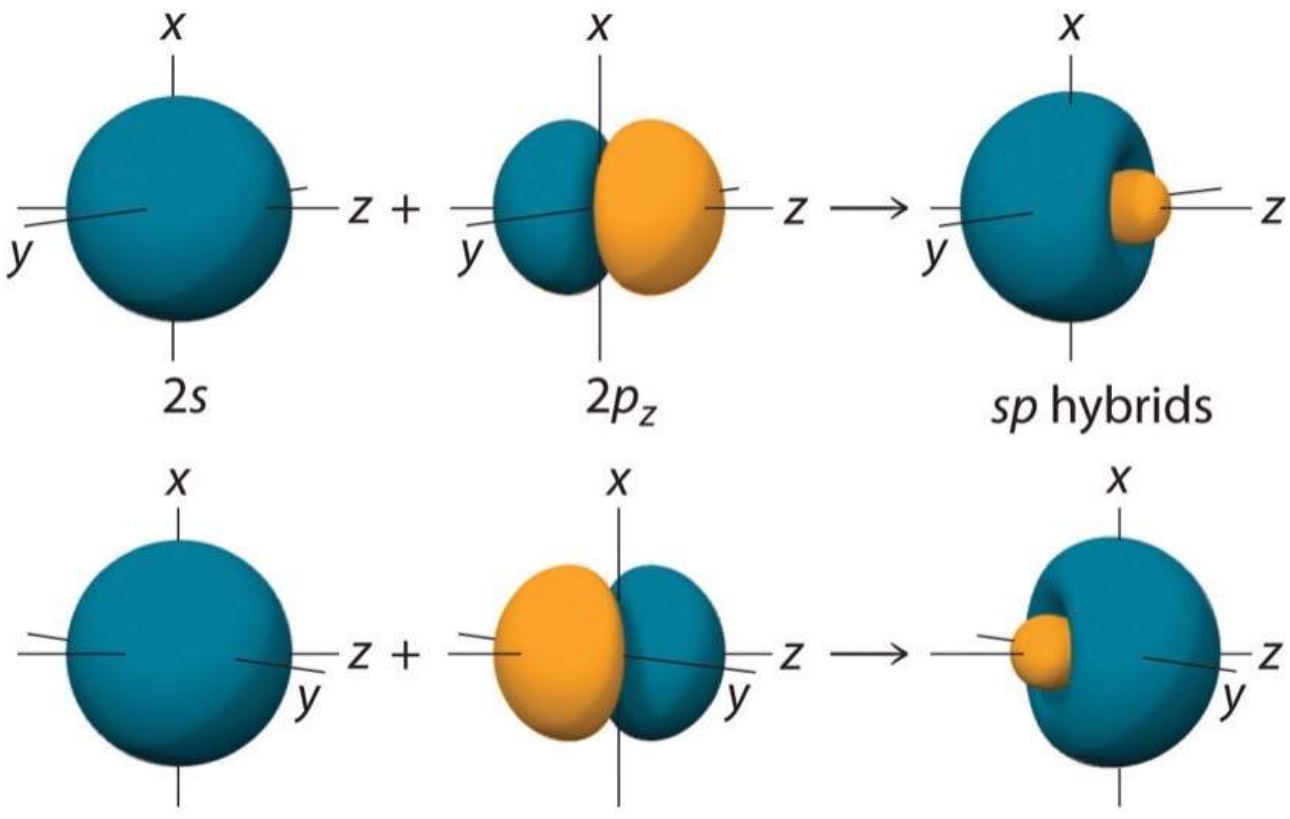
Examples of hybridisation:

[i]. sp hybridisation:

$$sp = \frac{1}{\sqrt{2}}(2s + 2p_z)$$

and

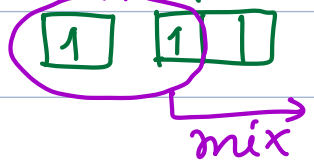
$$sp = \frac{1}{\sqrt{2}}(2s - 2p_z)$$



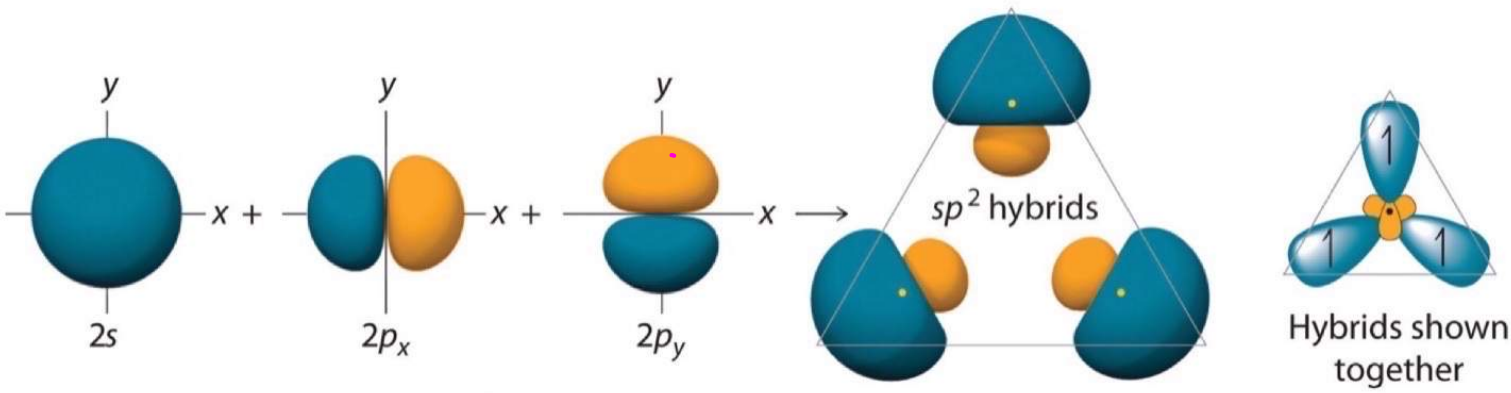
ex: BeH_2

Be: $2s^2\ 2p^0$

Be(excited) = $2s^1\ 2p^1$



[ii]. sp^2 hybridisation:

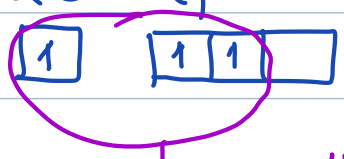


Formation of sp^2 Hybrid Orbitals. Combining one *ns* and two *np* atomic orbitals gives three equivalent sp^2 hybrid orbitals in a trigonal planar arrangement; that is, oriented at 120° to one another.

ex: BF_3

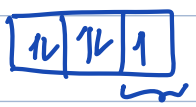
B: $2s^2\ 2p^1$

B (excited): $2s^1\ 2p^2$

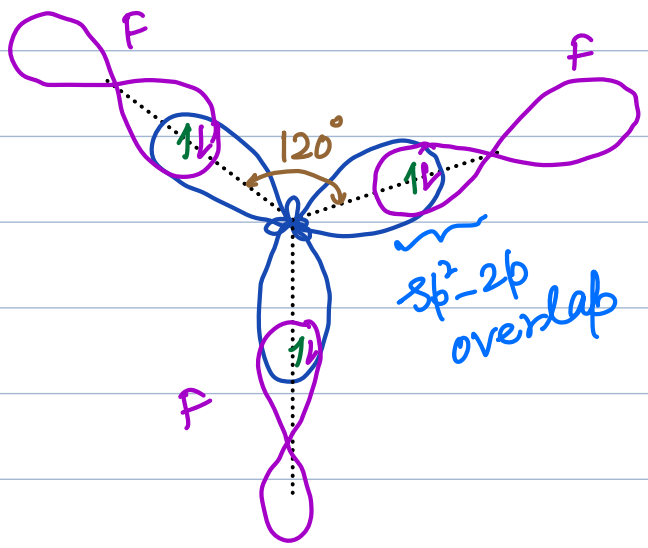


mix → three 'sp^2' orbitals

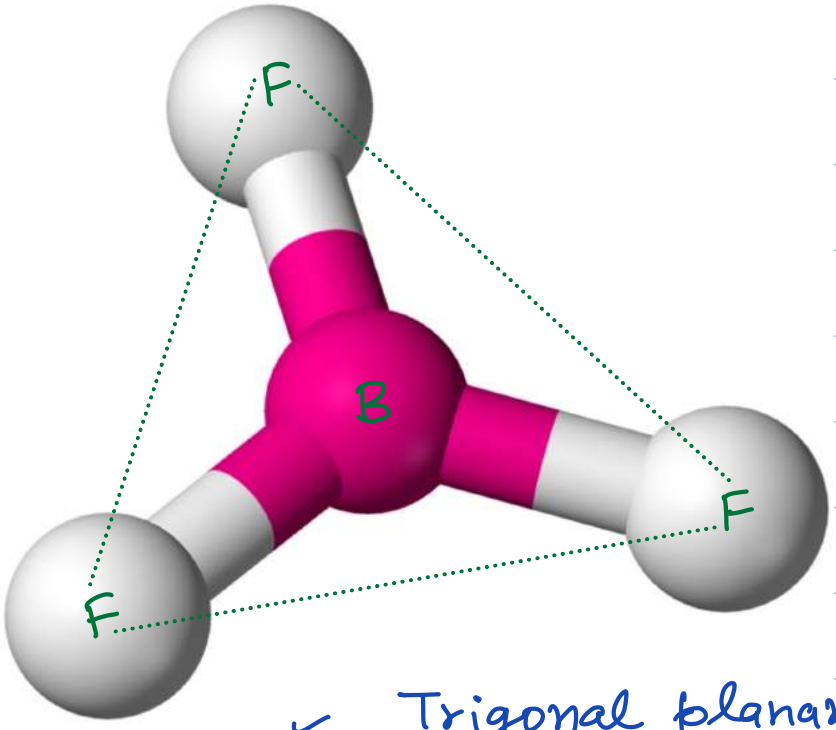
F: $2s^2\ 2p^5$



2p



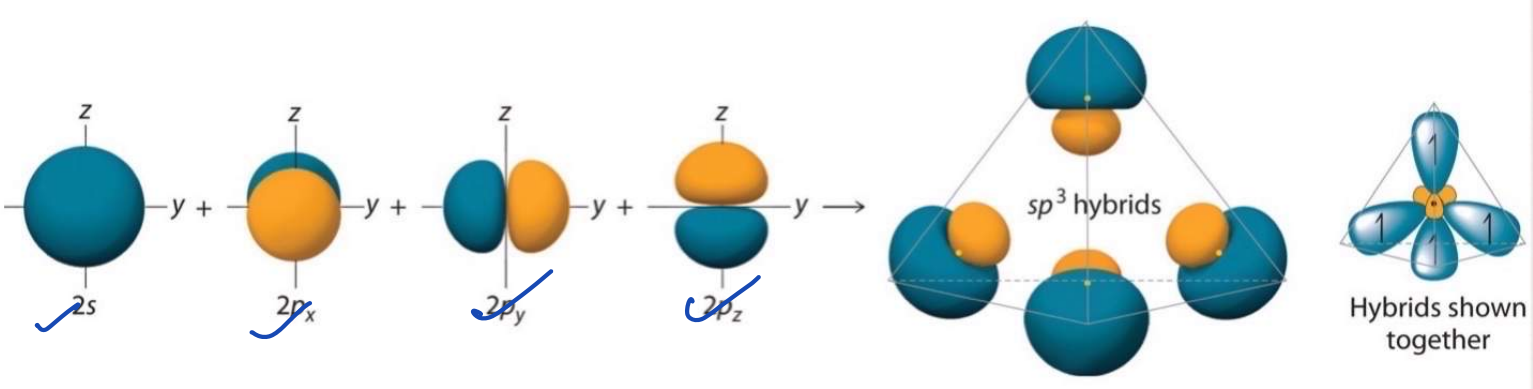
Bond angle = 120°



✓ Trigonal planar Geometry

✓ Triangular planar Geometry

[iii]. sp^3 hybridisation :

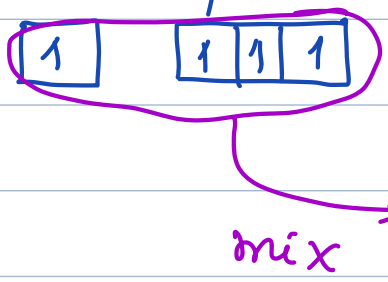


Formation of sp^3 Hybrid Orbitals. Combining one *ns* and three *np* atomic orbitals results in four sp^3 hybrid orbitals oriented at 109.5° to one another in a tetrahedral arrangement.

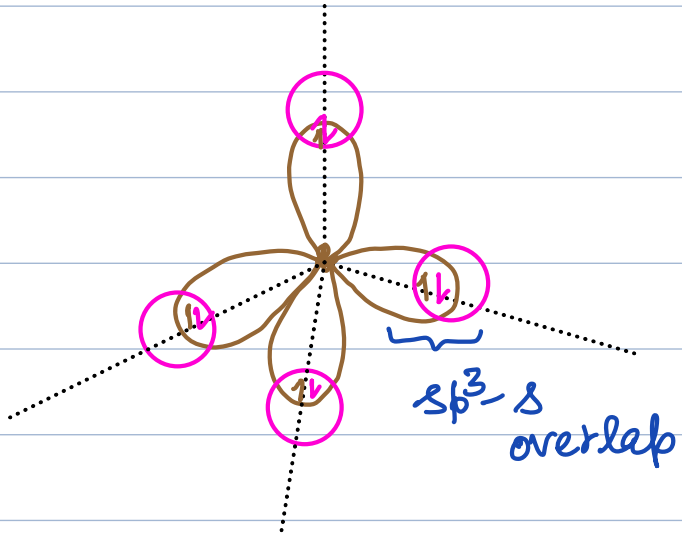
ex: CH_4

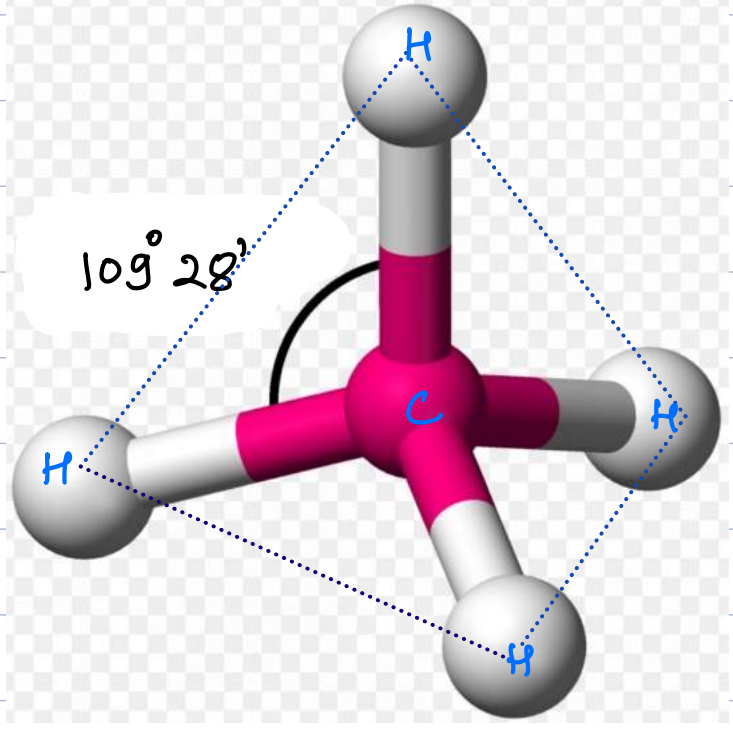
H: $1s^1$

C: $2s^1$ $2p^3$



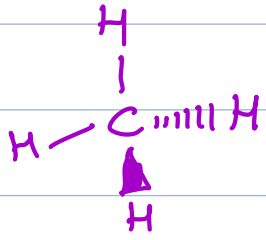
mix → four sp^3 orbital



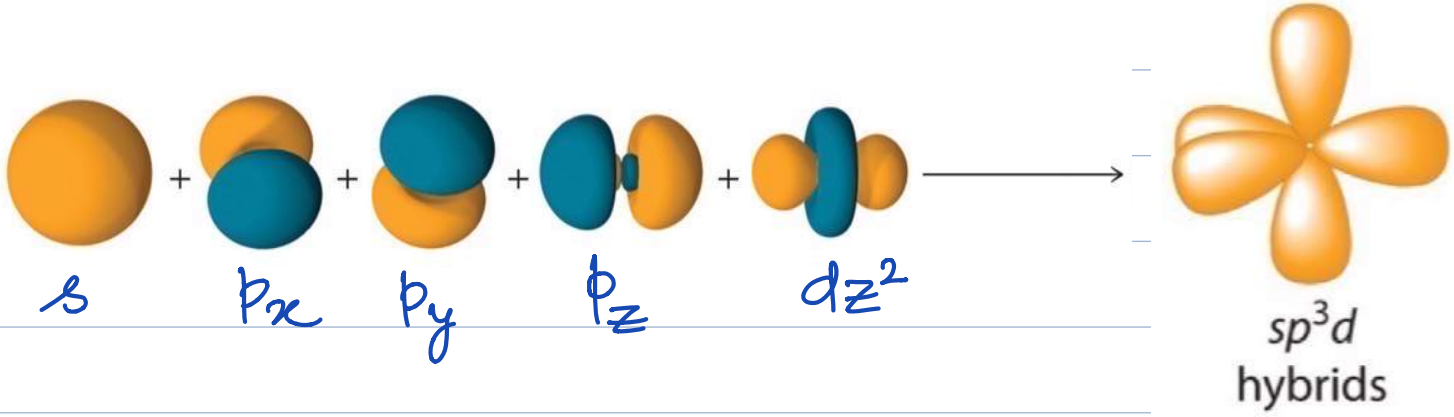


≈ Tetrahedral geometry

≈ Bond angle = 109° 28'



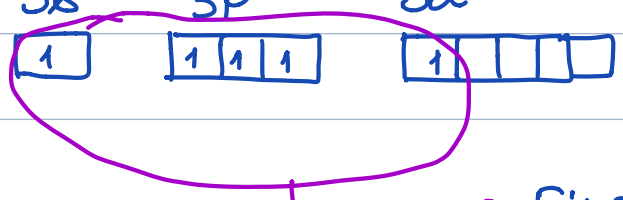
[iv]. sp^3d hybridisation :



ex: PCl_5

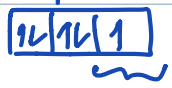
P: $3s^2\ 3p^3\ 3d^0$

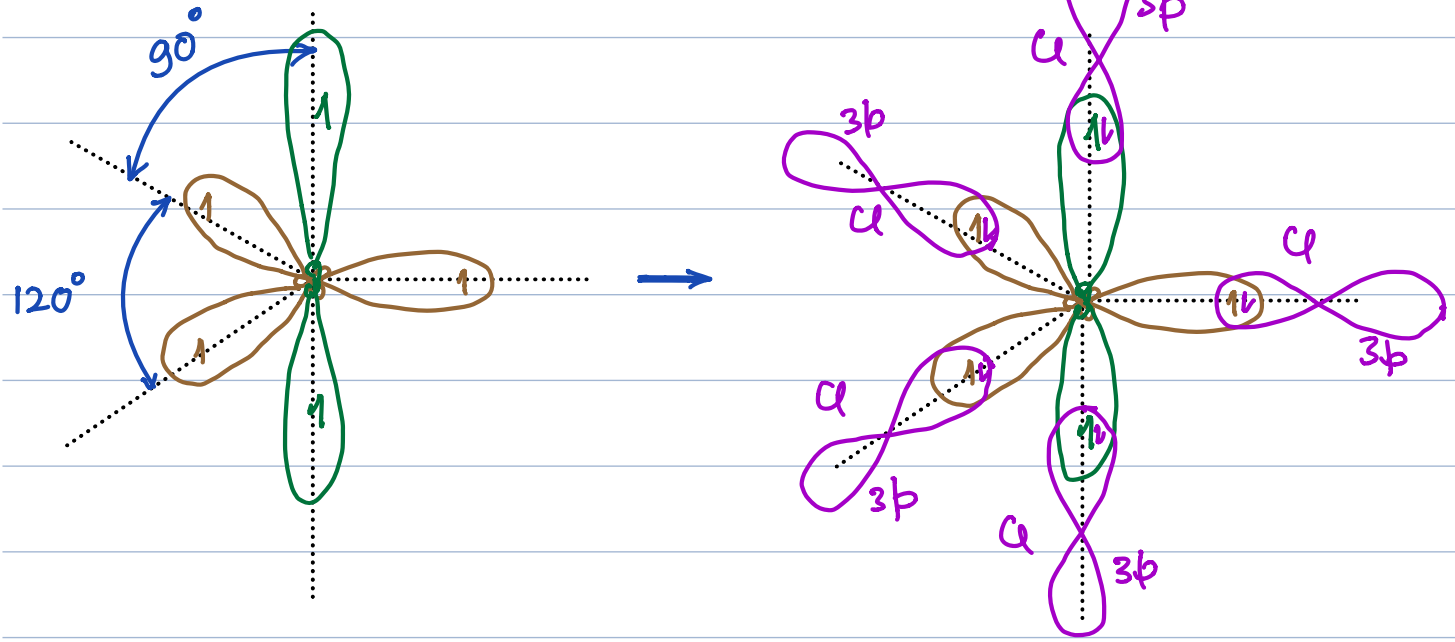
P: (excited) $3s^1\ 3p^3\ 3d^1$

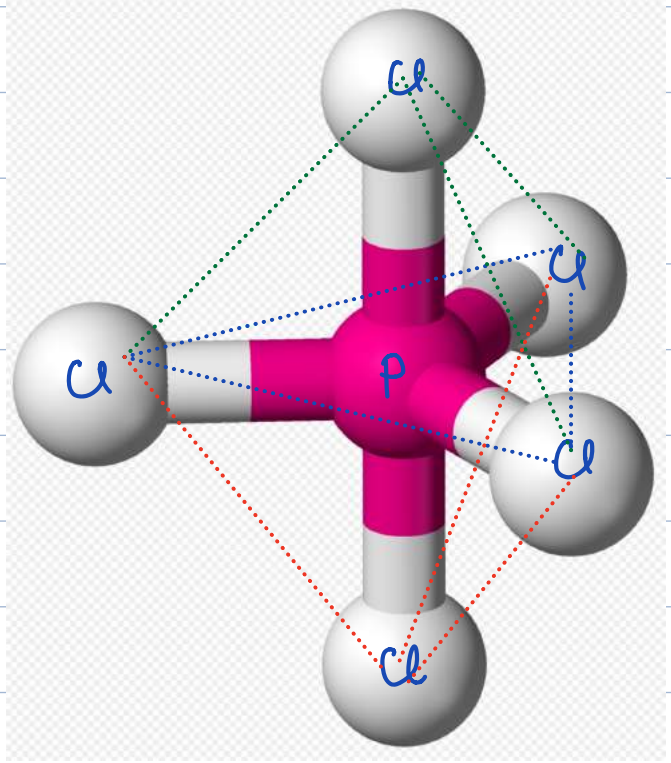


mix → Five 'sp^3d' orbitals

Cl: $3s^2\ 3p^5$

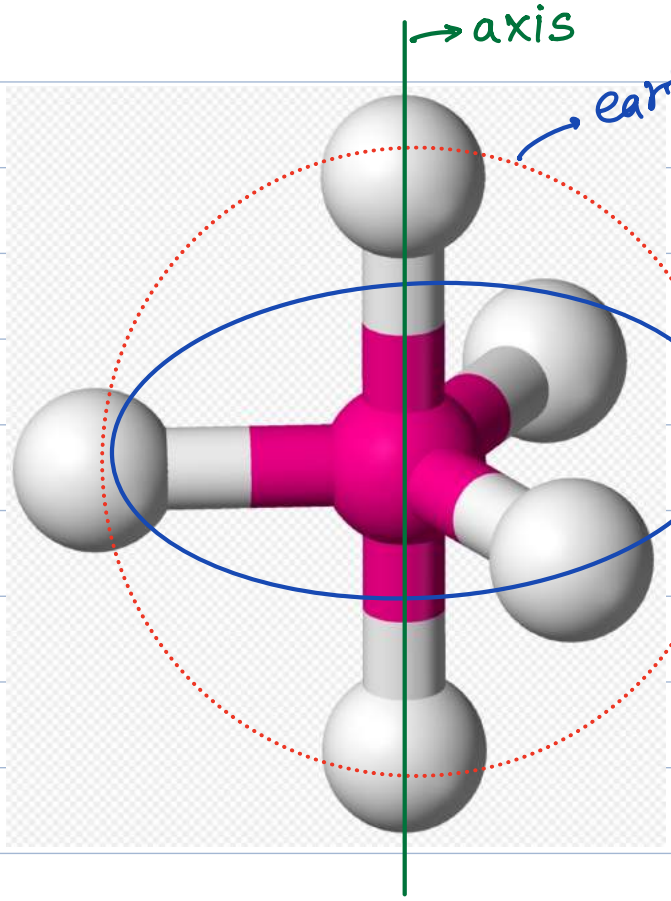


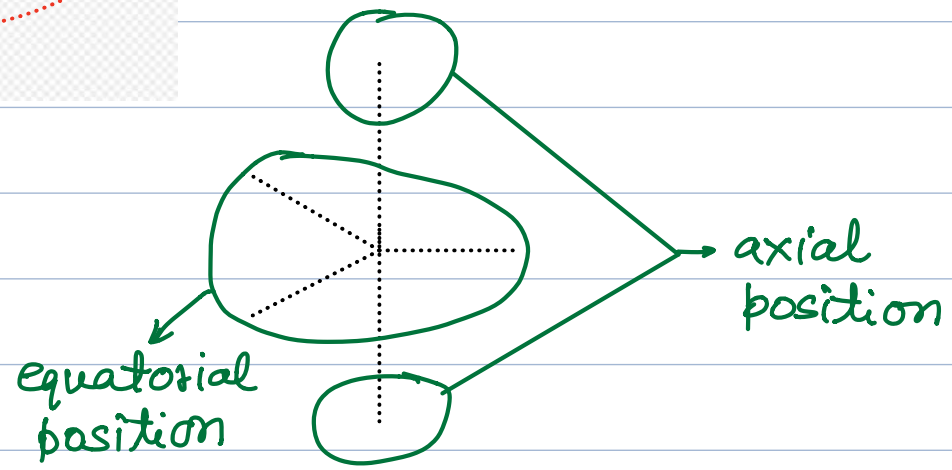




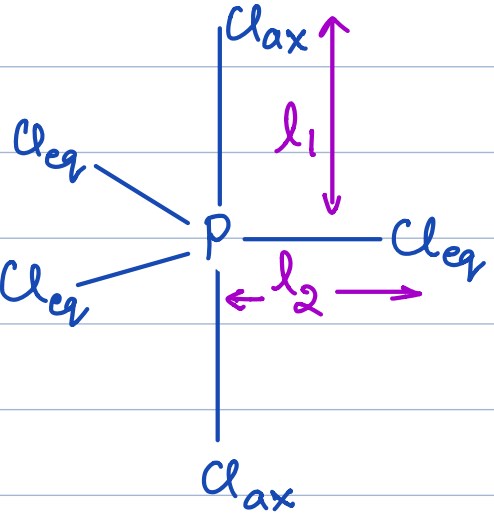
Geometry: Trigonal bipyramidal (TBP)

Bond angles: 120°, 90°





Imp.



✓ $[l_1 > l_2]$

axial bond length is more than equatorial bond length; ∵ axial orbitals face greater repulsion.

So,

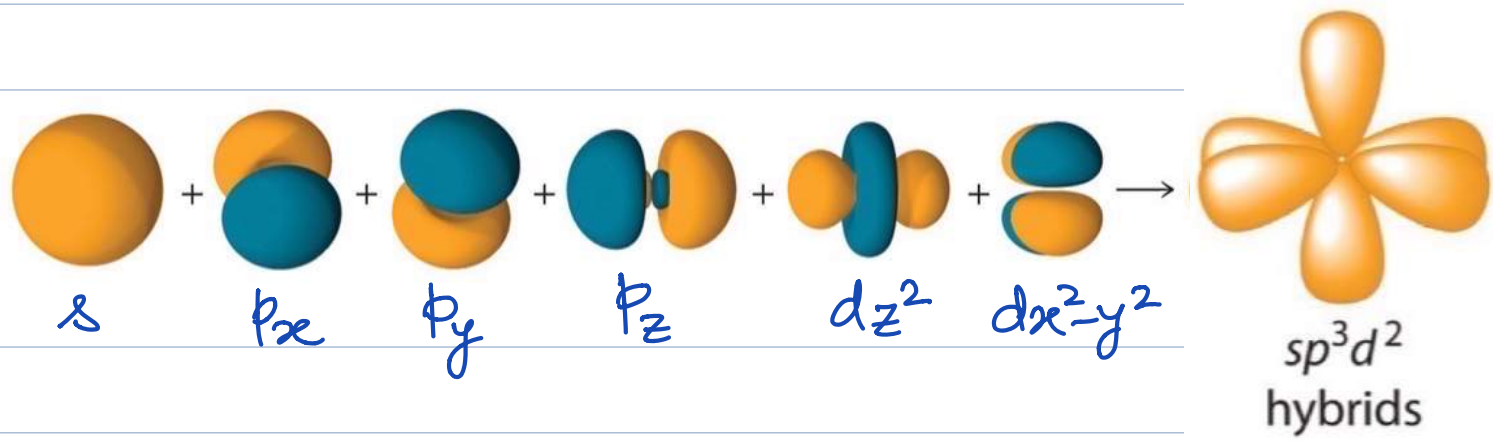
$$PCl_5 \xrightarrow{\Delta} PCl_3 + Cl_2$$

✓ In PCl_5, all five hybrid orbitals are not of equal size.

equatorial	3 - orbitals:	same size
Axial	2- orbitals:	same size (larger than equatorial)

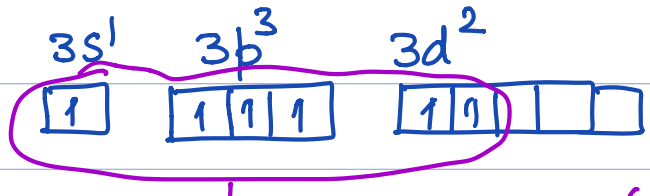
- actually, sp^3d hybridisation is:
  - ✓ $(sp^2 + pd)$ hybridisation

[v]. sp^3d^2 hybridisation:

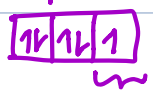


ex: SF_6

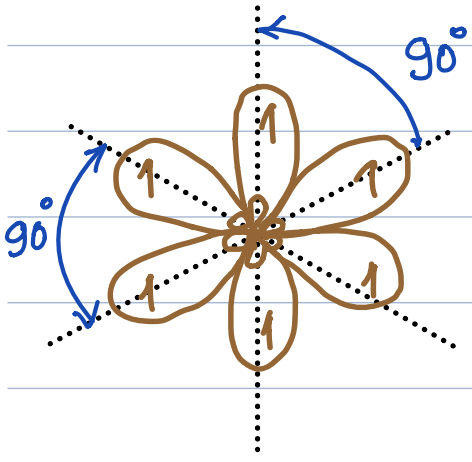
S: $3s^1$ $3p^3$ $3d^2$

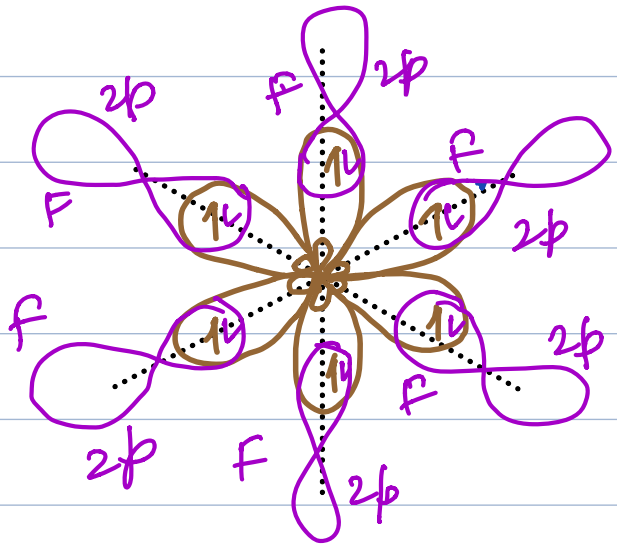


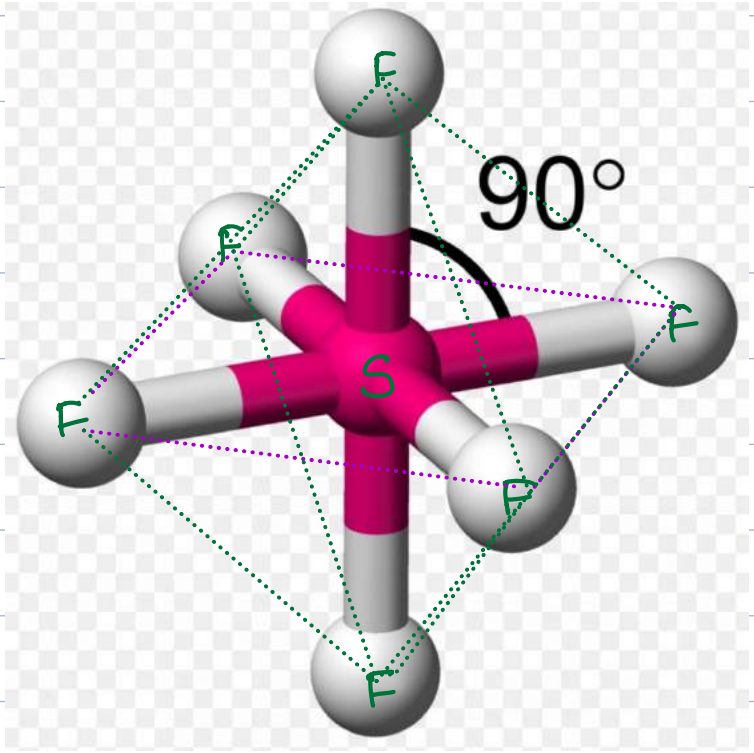
F: $2s^2$ $2p^5$



mix → six 'sp^3d^2' orbitals



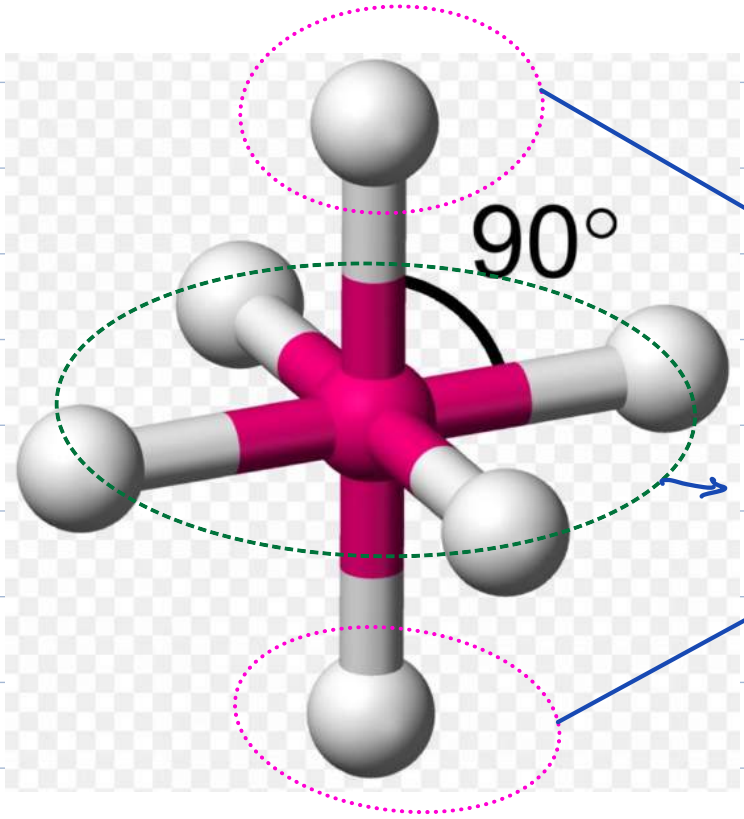




Geometry: square bipyramidal (or) octahedral

Bond angle: 90°

∵ All bonds are of equal length

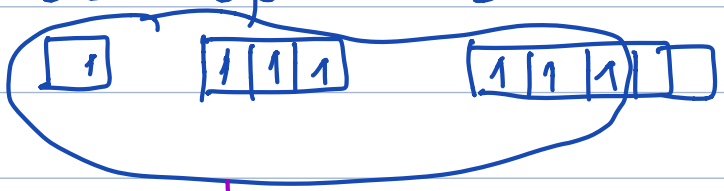


[vi]. sp^3d^3 hybridisation:

$(s + p_x + p_y + d_{x^2-y^2} + d_{xy}) + (p_z + d_{z^2})$

ex: IF_7

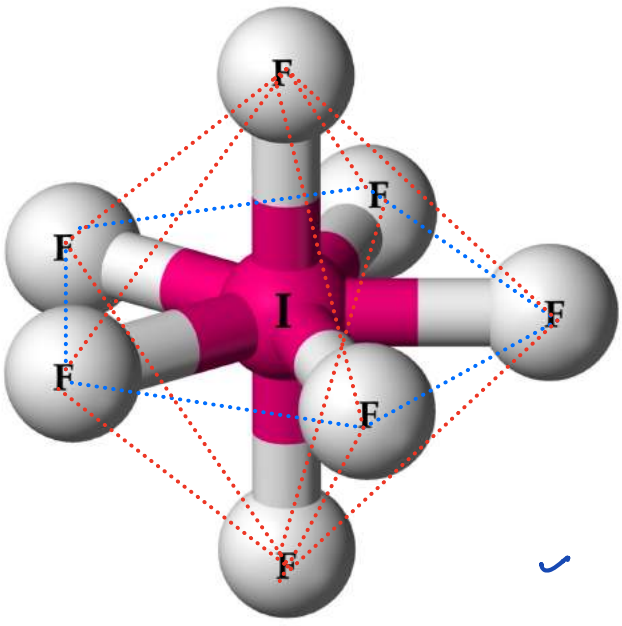
I : $5s^1$ $5p^3$ $5d^3$

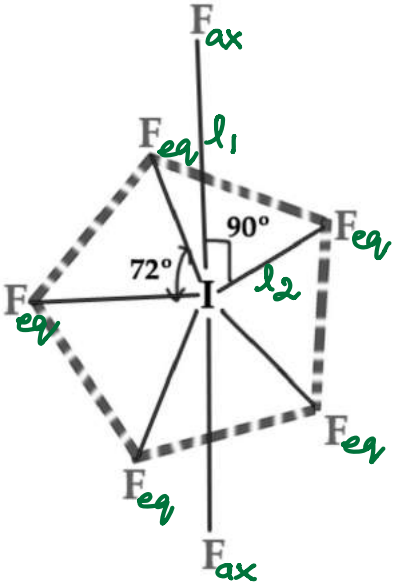


mix → seven 'sp^3d^3' orbitals

Geometry: Pentagonal bipyramidal (PBP)

Bond angles: 72°, 90°





- $l_2 > l_1$

equatorial orbitals face more repulsion than axial.

Finding hybridisation on central atom:

No. of hybrid orbitals = No. of σ bonds + No. of LP

(i) $\ddot{N}H_3$

σ = 3, LP = 1 → sum → 4

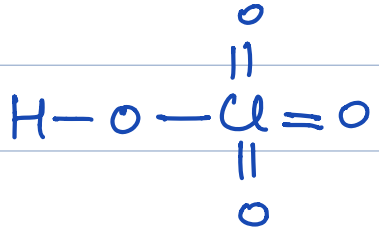
sp^3 hybridisation on N

(ii) $ICl_2^{\ominus}$

σ = 2, LP = 3 → sum → 5

sp^3d hybridisation on I

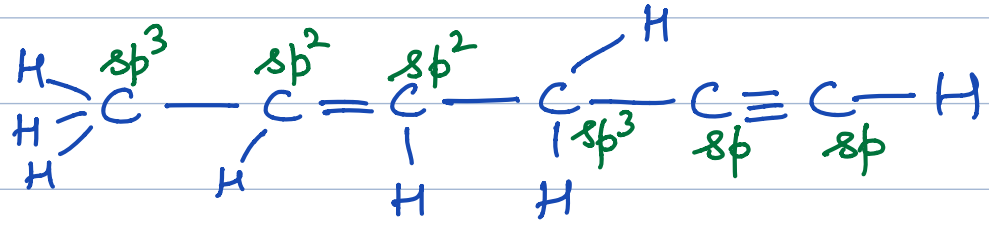
(iii) $HClO_4$



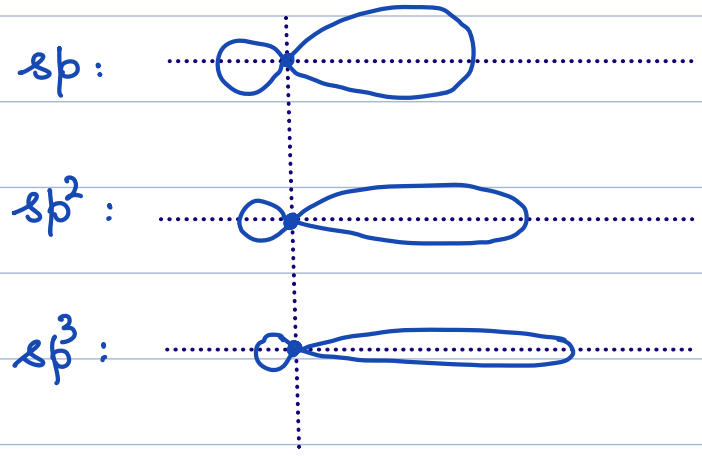
σ = 4, LP = 0 → sum → 4

sp^3 hybridisation on Cl

(iv)



Hybridisation	% s-character	% p-character
sp	50 %	50 %
sp^2	33.3 %	66.7 %
sp^3	25 %	75 %



inc. % p-character
inc. length of orbital
dec. thickness/bulkiness of orbital

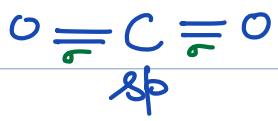
More the % s-character, more will be the EN of element. (for same element)

✓ inc. % s character inc.

✓ inc. EN of C

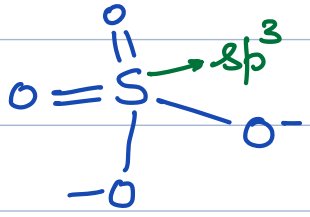
Finding type of π-bond in given species:

(i) CO_2 :



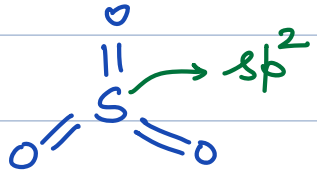
✓ two $p\pi - p\pi$ bonds

(ii) SO_4^{2-} :



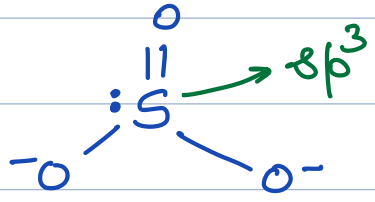
S → d
O → p
→ Two $p\pi - d\pi$ bonds

(iii) SO_3 :



S → one 'p'
two 'd'
O → p
→ one $3p\pi - 2p\pi$ bond + Two $3d\pi - 2p\pi$ bond

(iv) SO_3^{2-} :



S: 3d
O: 2p
→ one $d\pi - p\pi$ bond

Valence shell electron pair repulsion theory:

(VSEPR Theory)

- The shape of a molecule depends upon the number of valence shell electron pairs (bonded or nonbonded) around the central atom.
- Pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged.
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.
- The valence shell is taken as a sphere with the electron pairs localising on the spherical surface at maximum distance from one another.
- A multiple bond (single bond of) is treated as if it is a single electron pair and the two or three electron pairs of a multiple bond are treated as a single super pair. (conclusion: ≡ bond give more repulsion than = bond which give more repulsion than — bond).
- Where two or more resonance structures can represent a molecule, the VSEPR model is applicable to any such structure.

* A Lone pair occupies more space than a bond pair around central atom. (∵ LP attracted by one nucleus while bond pair is attracted by more than one nucleus)

[LP: lone pair , BP: bond pair]

* one LP – one LP > one LP – one BP > one BP – one BP

inc. repulsion

- LP जहाँ → angle ज्यादा जहाँ
- more EN surrounding जहाँ → angle कम जहाँ

Geometry / Shape : LP is not considered.

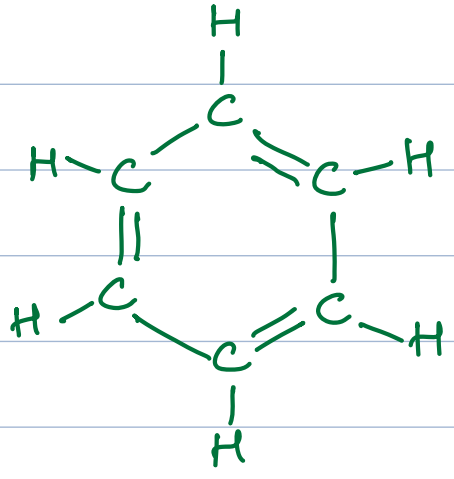
electron geometry : LP is also considered

isostructural : means same shape

Resonance :

C–C bond length = 154 pm
C=C bond length = 134 pm

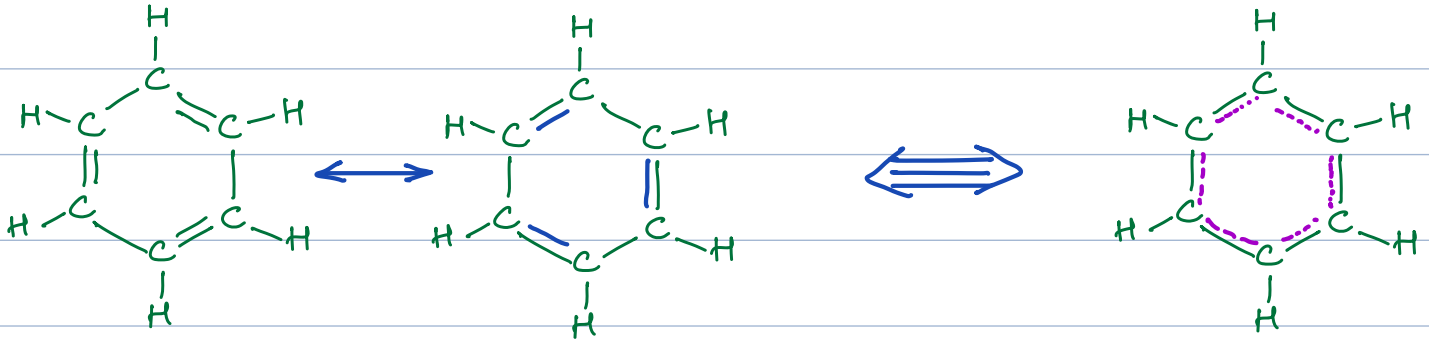
Benzene (C_6H_6) : experimentally obtained data
all carbon–carbon bond lengths are equal and value is 139 pm.



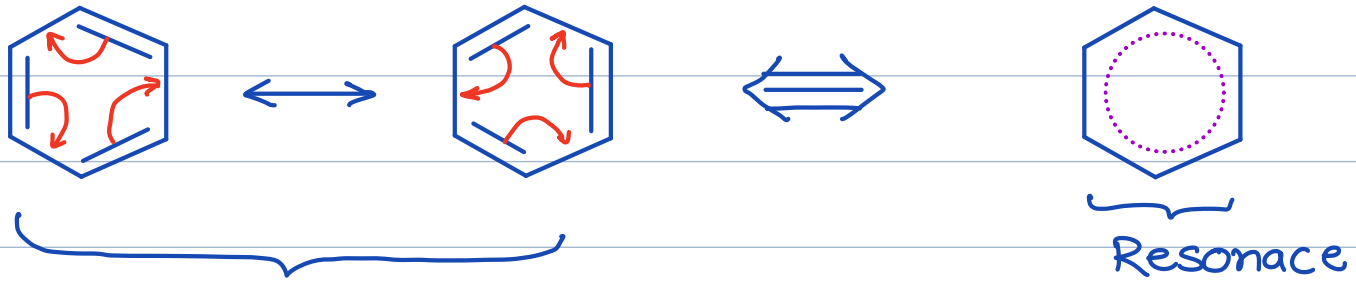
Acc. to this structure, there should be two different bond lengths b/w carbon and carbon (ie. one value acc to single bond & other value acc. to double bond.

✓ so a single Lewis structure of benzene can't explain the bond length data.

Definition : The phenomenon in which a species can be represented by two or more Lewis structures (none of which can alone explain all the properties of species) having same arrangement of atoms but different distribution of electrons is called "Resonance".



Structure - 1 Structure - 2 Resonance hybrid



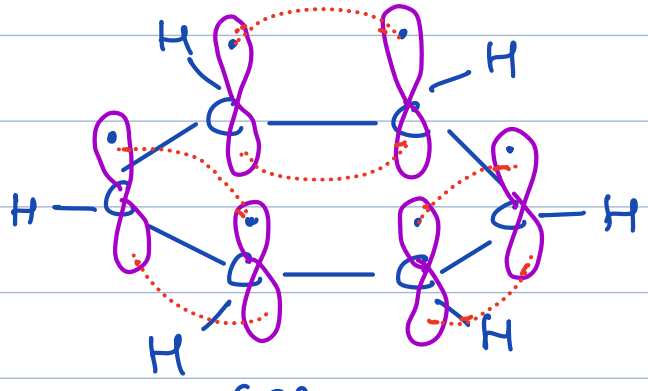
Resonating structures
(OR)
Contributing structures
(OR)
Cannonical structures

Resonace hybrid

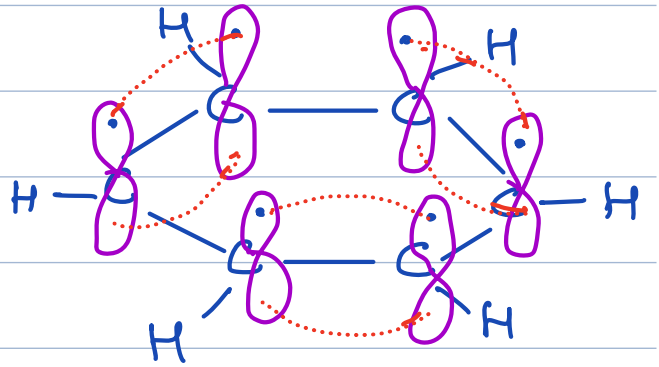
- Resonance arises due to delocalisation of electrons of π-bond or lone pair.
- Resonating structures do not exist in reality, they are on paper only.
- Resonance hybrid structure is the most close to actual structure of species in properties.

Conditions of Resonance :

[1]. The part of species or the species as a whole where resonance is taking place must be planar or nearly so.



(Planar)



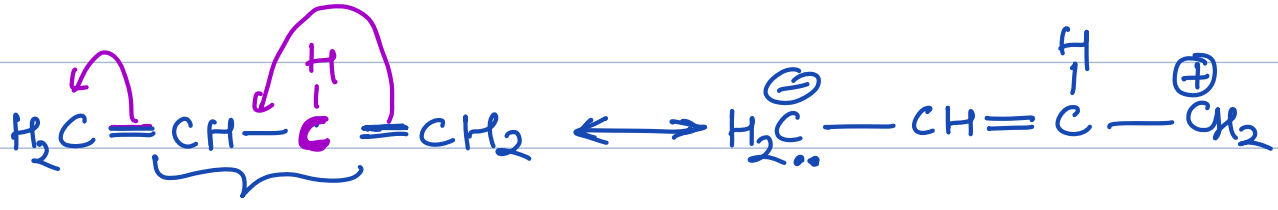
(Planar)

[2]. Presence of conjugation :

(A) π- π conjugation :

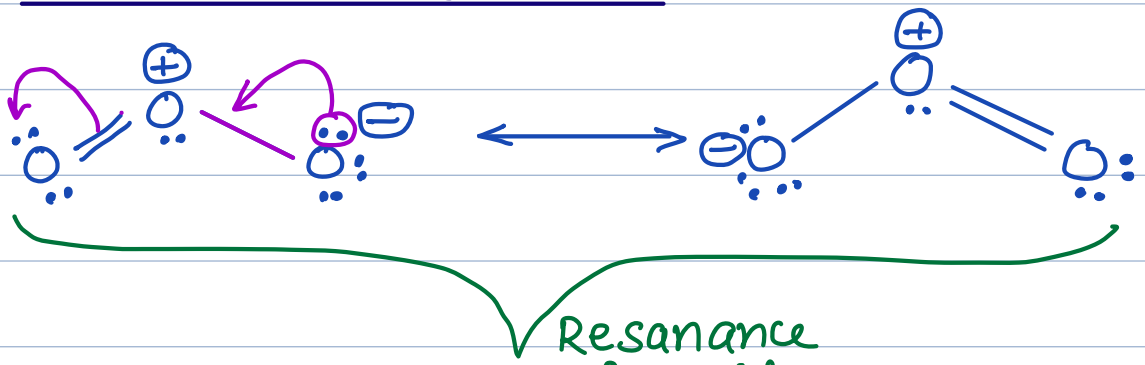
✓ presence of alternate π- bonds

ex:

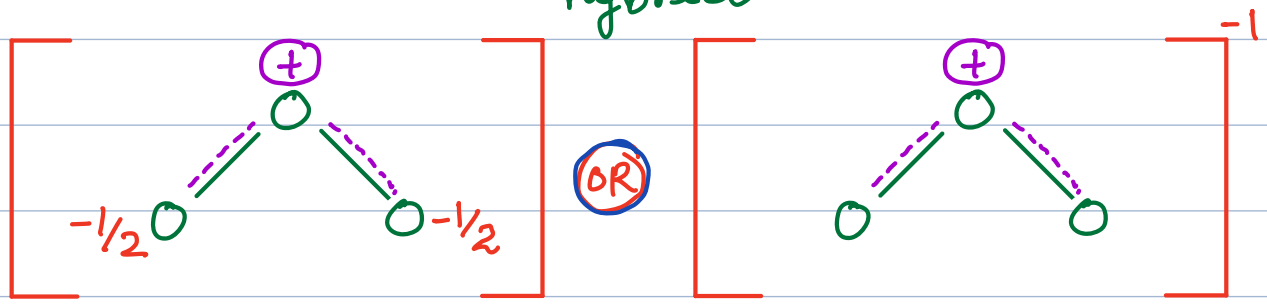


Other ex: Benzene

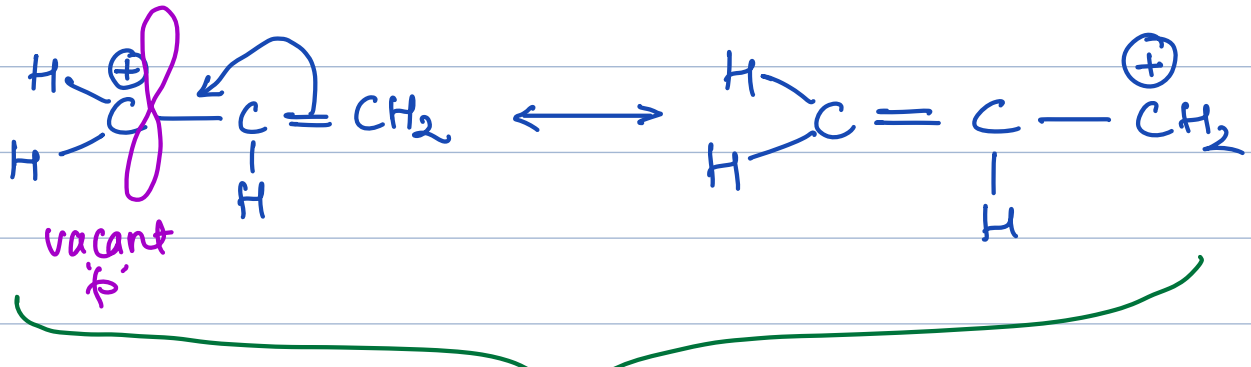
(B) LP – π conjugation:



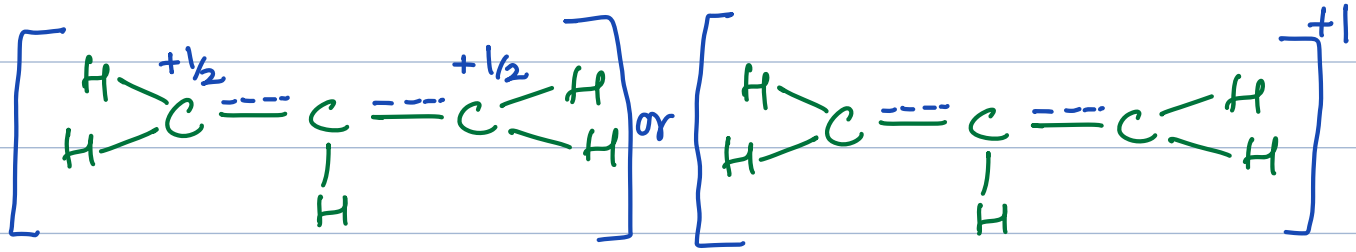
Resanance hybrid



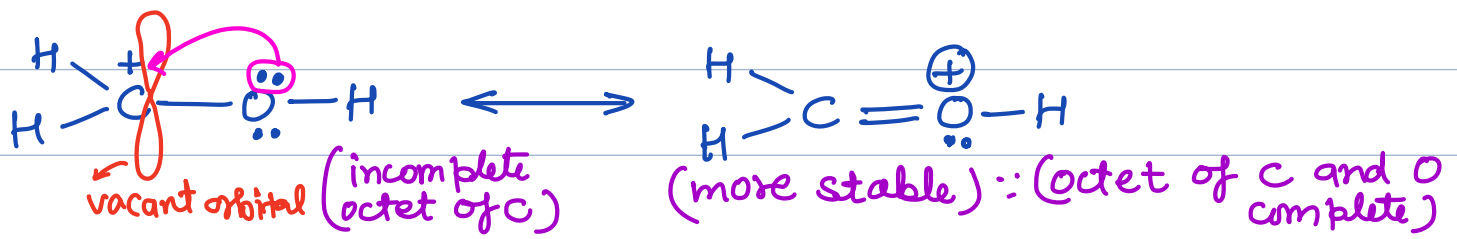
[C]. vacant orbital – π conjugation:



Resonance hybrid

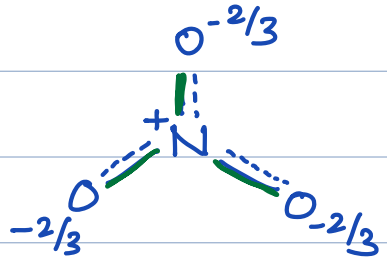


[D] vacant orbital – LP conjugation:



Ques. Does all hybrid orbitals of NO_3^- are equivalent?

Sol:



- All three hybrid orbitals of N are equivalent.

Do not use single resonating structure.

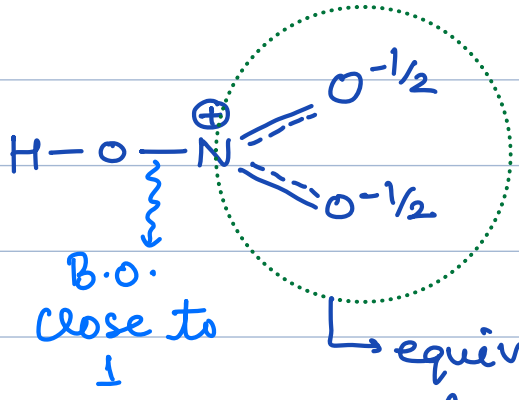
Bond parameters:

(i) Bond length
(ii) Bond angle
(iii) Bond dissociation energy

(i) **Bond length:** Internuclear distance b/w two bonded atoms.

✓ Bond length is affected by:

(a) size of surrounding atom
(b) Bond order
(c) Resonance
(d) % s-character (or) % p-character



B.O. close to 1

H is the smallest atom. So shortest bonds are those formed by the hydrogen.

equivalent

$l_3 = l_4$

(B.O. = 1.5)

Negligible contribution

Highly unstable ∵ +ve charge on adjacent atom & +ve on highly EN O

[ii] Bond angle: Angle between two covalent bonds is bond angle

Prioritywise rules for comparing bond angle:

(A). { no LP on central atom
all surrounding same and identically attached
→ exact bond angle as expected from hybridisation (no matter how EN are central and surrounding)

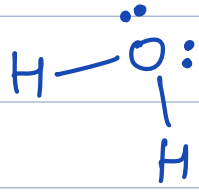
ex: $\overset{sp^3}{CH_4} = \overset{sp^3}{SiH_4} = \overset{sp^3}{GeH_4} = \overset{sp^3}{CCl_4} = \overset{sp^3}{SiCl_4} = 109°28'$

$\overset{sp^2}{BF_3} = \overset{sp^2}{BCl_3} = \overset{sp^2}{BBr_3} = \overset{sp^2}{BI_3} = 120°$

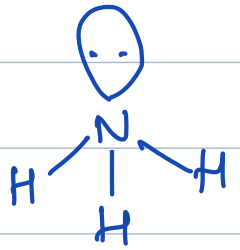
$F-\underset{sp}{Be}-F = Cl-\underset{sp}{Be}-Cl = 180°$

(B). Presence of lone pair on central atom may decrease the bond angle.

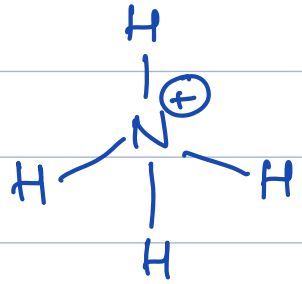
ex:



Bond angle = 104.5°



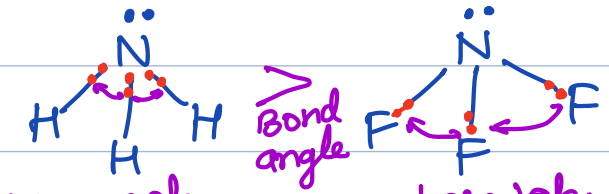
Bond angle less than 109° 28'

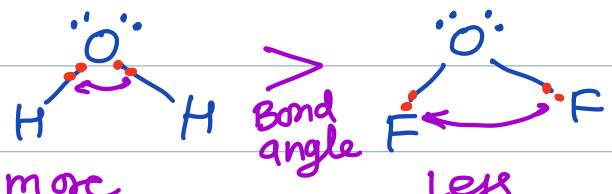


Bond angle = 109° 28'

(C). The presence of more electronegative surrounding atom usually decreases the bond angle.
(central atom must have equal no. LP, same CA)

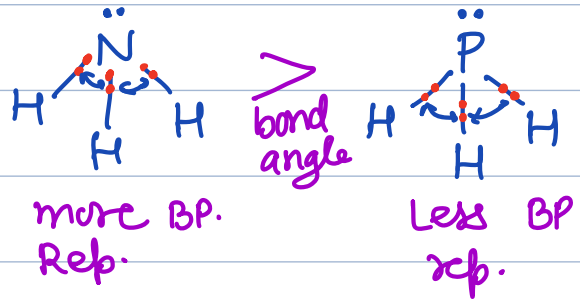
ex:





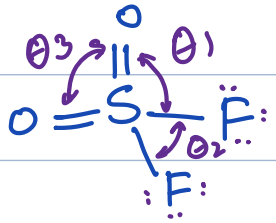
(D). If all surrounding atom are identical in species being compared & equal no. of LP on central atom, smaller bond angle in species having less EN central atom.

ex:

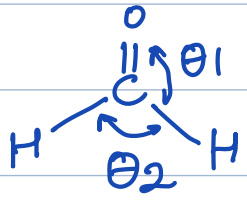


(E). Due to repulsion b/w multiple bonds (= or ≡) bond angle b/w them increases.

ex:



$[\theta_3 > \theta_1 > \theta_2]$



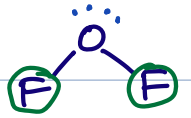
$[\theta_1 > \theta_2]$

steric repulsion

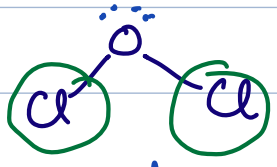
(F). Due to repulsion between bulky surrounding atoms or groups, bond angle increases.

ex. to learn:

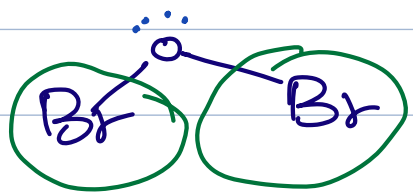
✓



Bond angle = 103°

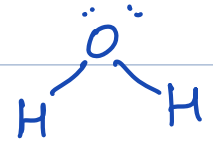


Bond angle = 111°

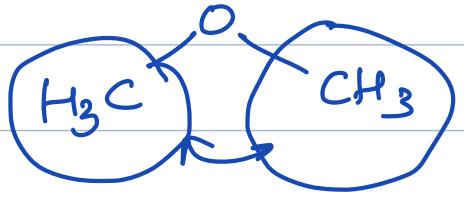


Bond angle = 112°

Bond angle > 109° 28'

≈ 

$= 104.5^{\circ}$



(C–O–C) Bond angle $> 109^{\circ}28'$

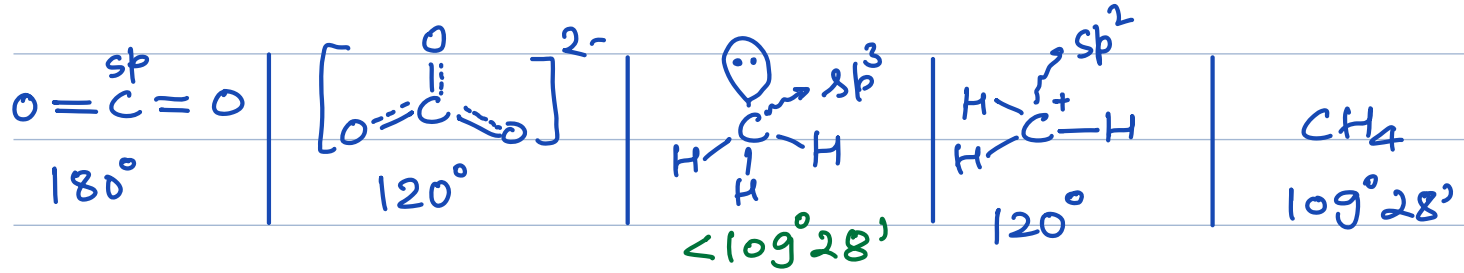
≈ $\underbrace{PF_3 < PCl_3 < PBr_3 < PI_3}_{\text{Bond angle}} < 109^{\circ}28'$

(on basis of steric repulsion)
or EN of surrounding

Question: Compare bond angles in

CO_2, CO_3^{2-}, $\overset{\ominus}{C}H_3$, $\overset{\oplus}{C}H_3$, CH_4

Sol:



$\underbrace{CO_2 > (CO_3^{2-} = \overset{+}{C}H_3) > CH_4 > \overset{\ominus}{C}H_3}_{\text{(Bond angle)}}$

Q: NH_3, NH_4^+, NH_2^-, NO_2^+, NO_3^- : compare bond angle

Sol: $\underset{180^{\circ}}{NO_2^+} > \underset{120^{\circ}}{NO_3^-} > \underset{109^{\circ}28'}{NH_4^+} > \ddot{N}H_3 > \ddot{N}H_2^-$

(iii) Bond dissociation energy:

(BDE) Energy needed to break 1 mole of bonds is bond dissociation energy.

BDE is affected by:

ⓐ Bond length
ⓑ Bond order
ⓒ LP–LP repulsions

BDE Comparisons:

(i) O_2 > O_3 > H_2O_2 (O–O BDE)

B.O. = 2 B.O. = 1.5 B.O. = 1

(ii) N–H > P–H > As–H > Sb–H

(iii) :F–F: < :Cl–Cl:

O–O < S–S

N–N < P–P

} on basis of LP–LP repulsion

(iv) O=O N=N N≡N

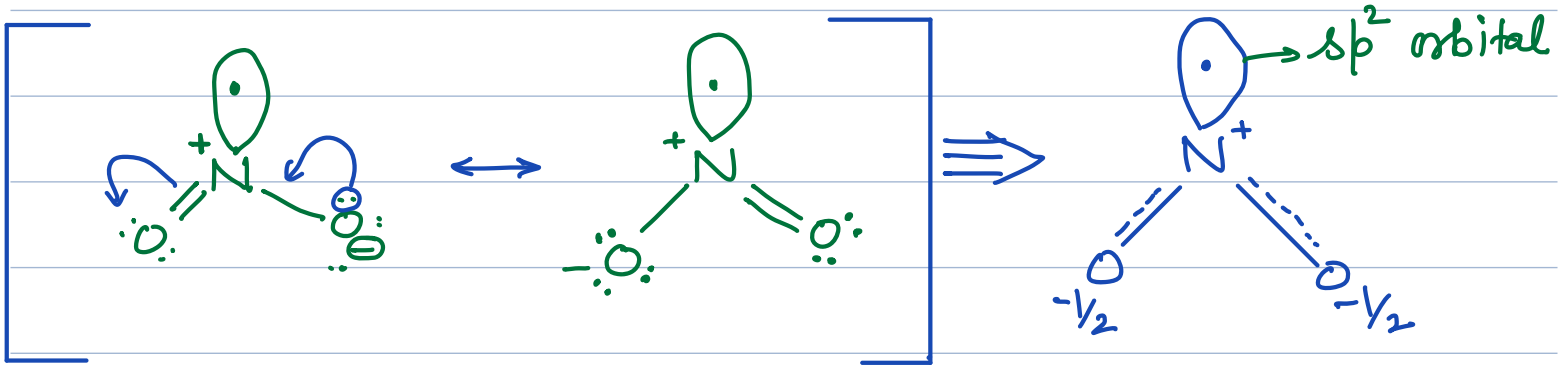
S=S P=P P≡P

} Bond length basis

odd electron species:

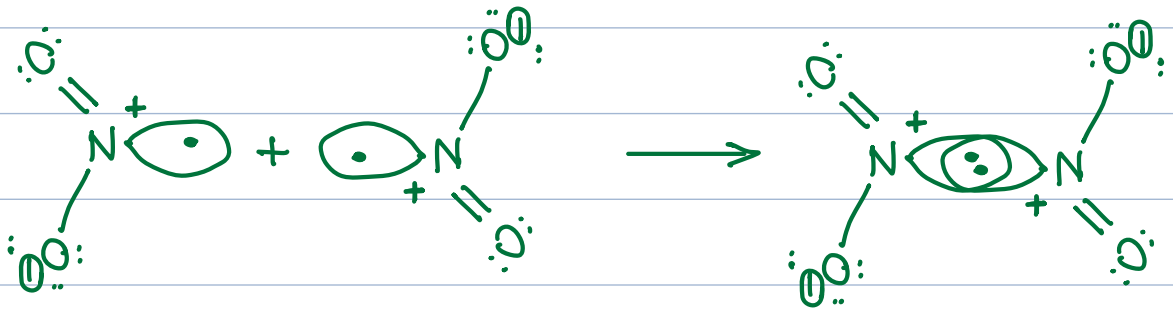
(i) NO_2 : Nitrogen dioxide : Reddish brown gas

(Paramagnetic species)



- Bent shape
- Bond angle > 120°
  (more BP repulsions)
  (= 134°)

At low temperature :

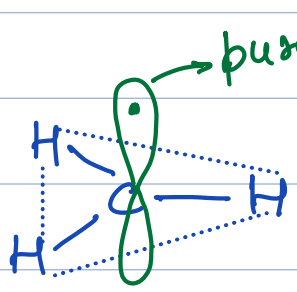


(N_2O_4)

(diamagnetic)

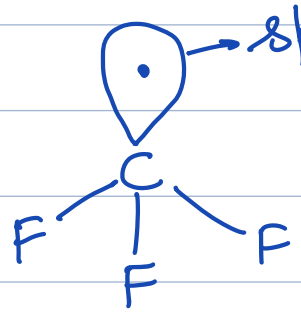
(ii) $\dot{C}H_3$: (Carbon free radical)

($2\,\dot{C}H_3 \longrightarrow C_2H_6$)



- Bond angle = 120°
- Trigonal planar

(iii). $\dot{C}F_3$

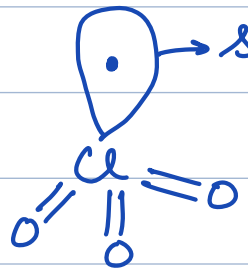


($2\,\dot{C}F_3 \longrightarrow C_2F_6$)

- Pyramidal shape
- Bond angle < 109°28'

(iv) $\dot{C}lO_3$

($2\,\dot{C}lO_3 \longrightarrow Cl_2O_6$)



(Pyramidal shape)

If surrounding atoms are more electronegative then odd e^- of central atom occupies hybrid orbital.

ClO_2 : Bent shaped
Bond angle $\simeq$ 118°
yellow-brown Paramagnetic gas

~~$2ClO_2 \longrightarrow Cl_2O_4$~~

Dipole moment (μ) :

$$|\mu| = |q| \text{ into } d$$

- It is defined as the product of the magnitude of the charge and the distance b/w centres of positive and negative charge.

- It is a vector quantity.

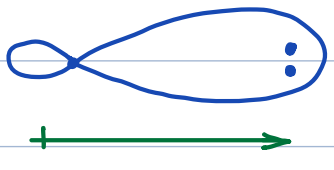
- In chemistry, presence of dipole moment is represented by the crossed arrow ($\longrightarrow$) put on correct shape of the species.

(cross on positive end, arrow head on negative end)

ex: $\overset{\delta+}{H}—\overset{\delta-}{Cl}$

$\longrightarrow$ showing direction of shift of e^- density.

- LP in a hybrid orbital

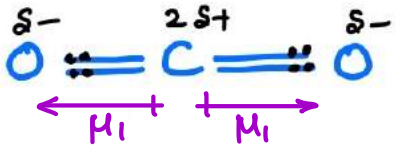


Net dipole moment (μ_{net}) = 0 → non-polar molecule

Net dipole moment (μ_{net}) ≠ 0 → Polar molecule

ex:

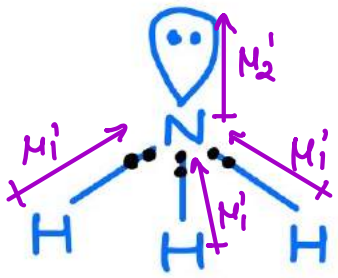
(i) CO_2



$\mu_{net} = 0$

(Non-polar)

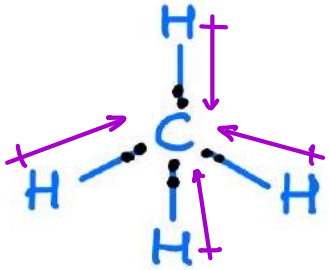
(ii) NH_3



$\mu_{net} \neq 0$

(Polar)

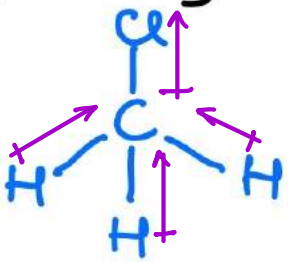
(iii) CH_4



$\mu_{net} = 0$

(Non-polar)

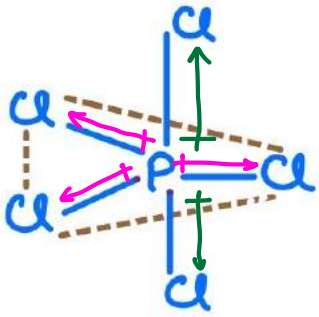
(iv) CH_3Cl



$\mu_{net} \neq 0$

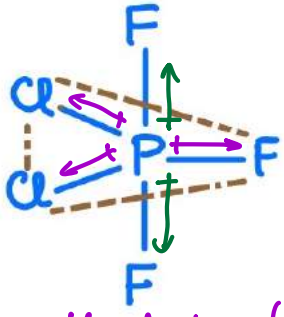
(Polar)

(v) PCl_5



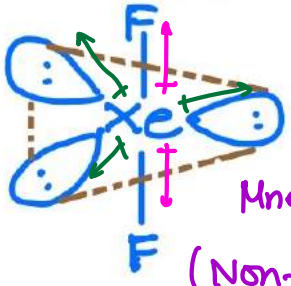
$\mu_{net} = 0$ (non-polar)

(vi) PCl_2F_3



$\mu_{net} \neq 0$ (Polar)

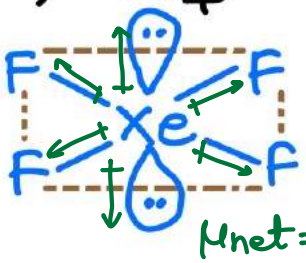
(vii) XeF_2



$\mu_{net} = 0$

(Non-polar)

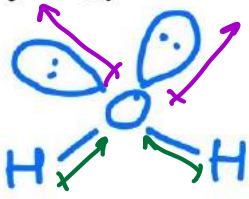
(viii) XeF_4



$\mu_{net} = 0$

(Non-polar)

(ix) H_2O

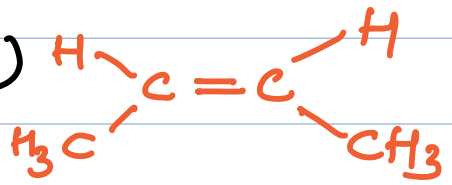


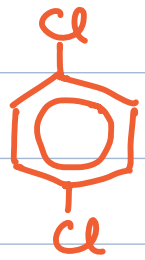
$\mu_{net} \neq 0$

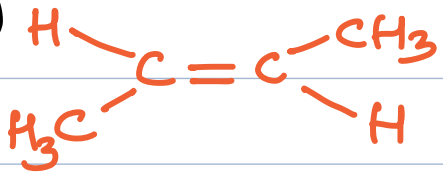
(Polar)

Q. which of the following species are non-polar?

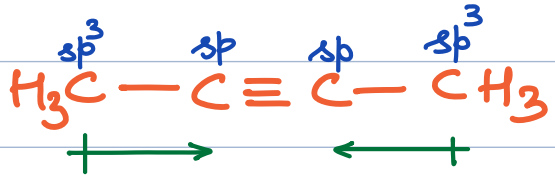
(A) $H_3C-C\equiv C-CH_3$

(B) 

(C) 

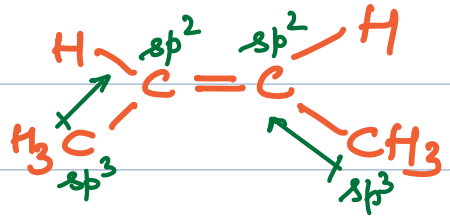
(D) 

Sol:

(A) 

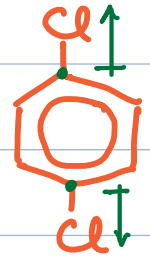
($\mu_{net} = 0$)

$(sp^3\ 'C' \underset{EN}{<} sp\ 'C')$

(B) 

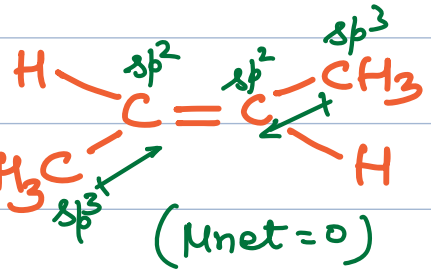
($\mu_{net} \neq 0$)

$[sp^2\ 'C' \underset{EN}{>} sp^3\ 'C']$

(C) 

($\mu_{net} = 0$)

$[C \underset{EN}{<} Cl]$

(D) 

($\mu_{net} = 0$)

$\underbrace{sp^2\ 'C' > sp^3\ 'C'}_{EN}$

Molecular orbital Theory (MOT):

Non existence of some species like He_2, Be_2, H_2^{2-}, etc. and paramagnetic behaviour of O_2, B_2 are explained by MOT.

Acc. to VBT:

$:\ddot{O}=\ddot{O}: \Rightarrow O_2$ should be diamagnetic but in reality, O_2 exhibits paramagnetic character.

Features of MOT:

(i) Electrons in a single atom are influenced by one nucleus (i.e. monocentric) while electrons in a molecule are influenced by two or more nuclei (i.e. polycentric).

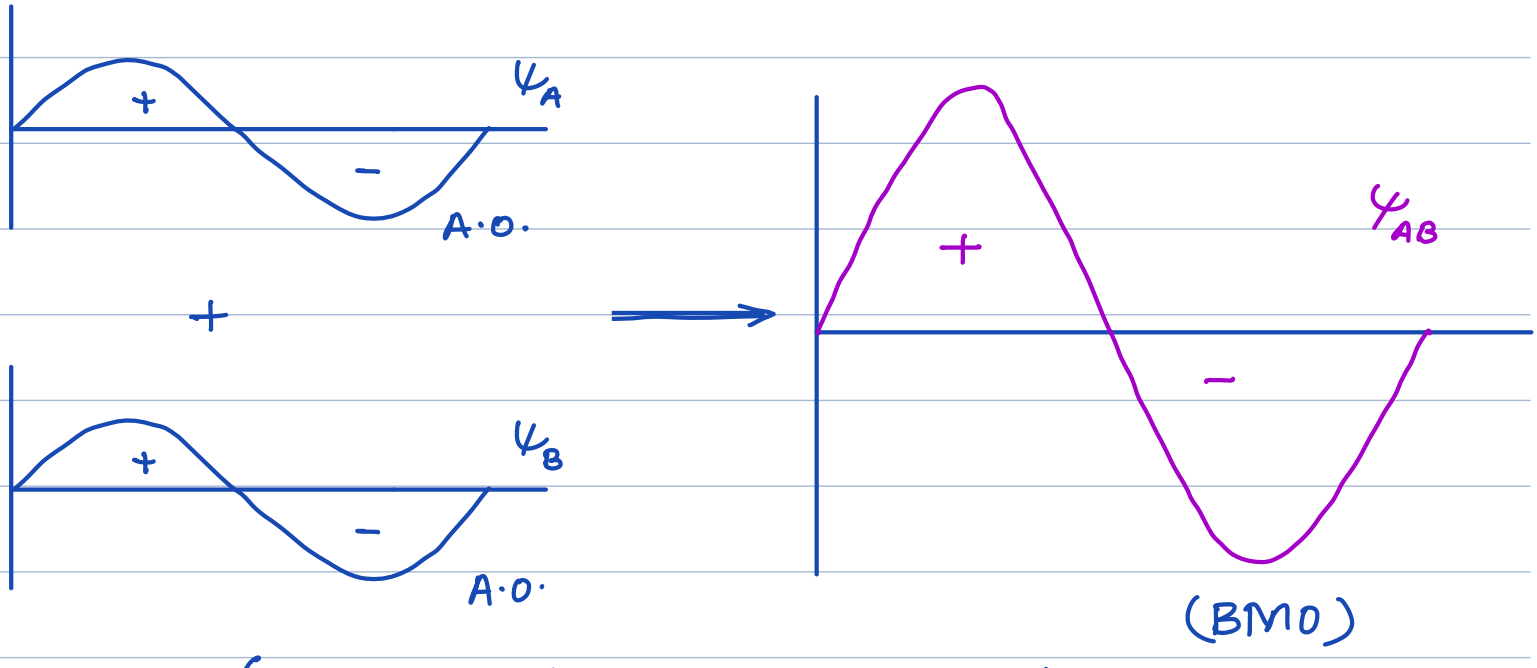
(ii). Like an atomic orbital, Molecular orbital is a three dimensional region of space where probability of finding electron is maximum.

(Atomic orbital: AO ; Molecular orbital: MO)

(iii). Atomic orbitals belonging to different atoms combine together to form the molecular orbitals.

*(iv). No. of atomic orbitals combined = No. of molecular orbitals formed

(v). When two atomic orbitals interfere with each other constructively, the electron probability density increases between nuclei and formed such molecular orbitals are called bonding molecular orbitals. (BMO)



(Constructive interference)

$$\psi_{AB} = \psi_A + \psi_B$$

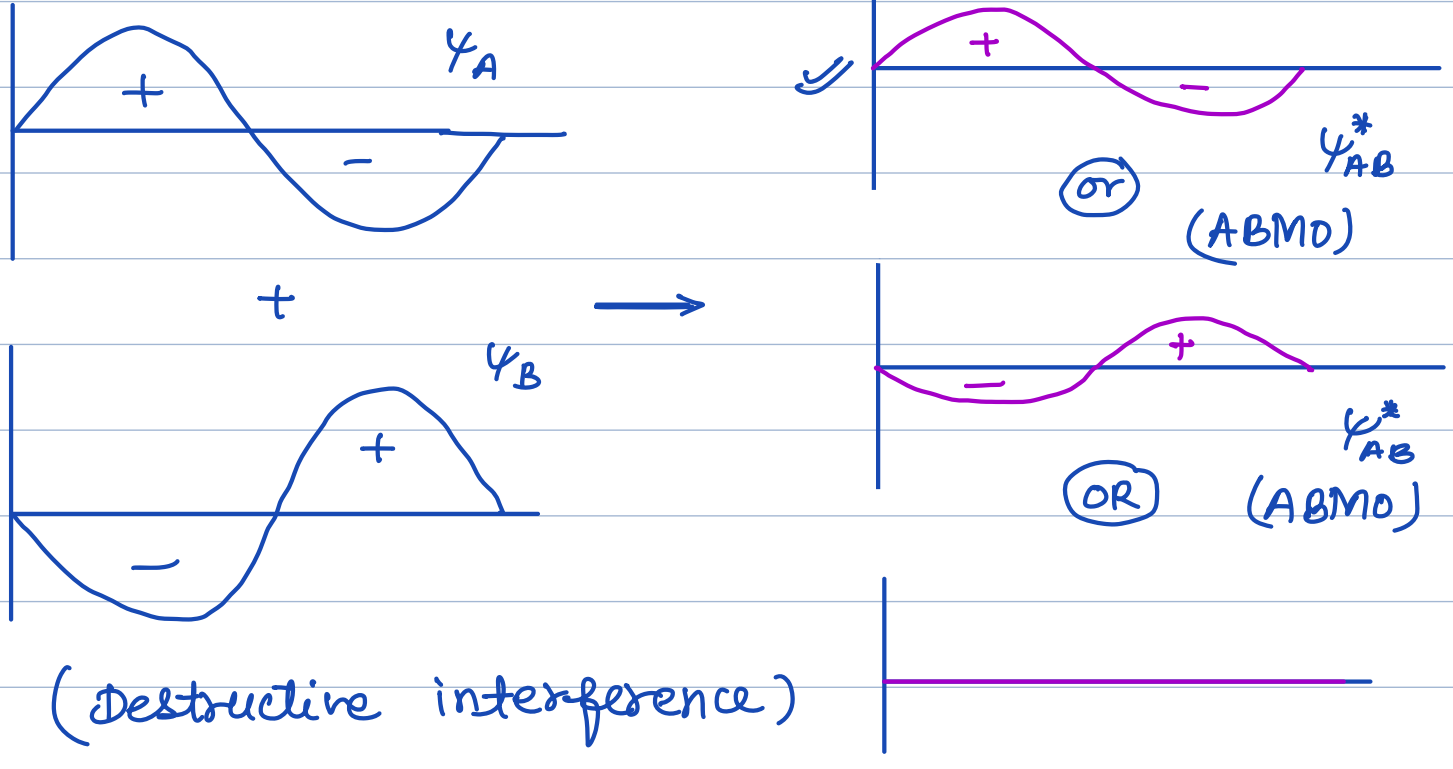
Squaring both sides

$$\psi_{AB}^2 = (\psi_A + \psi_B)^2$$

$$\left[\psi_{AB}^2 = \psi_A^2 + \psi_B^2 + 2\psi_A\psi_B \right]$$

Probability $\propto \psi_{AB}^2$

(vi) When two atomic orbitals interfere each other destructively, then electron probability density is decreased between nuclei. Such formed molecular orbitals are said to be antibonding molecular orbitals. (ABMO)



$$\psi^*_{AB} = \psi_A - \psi_B$$

squaring both sides,

$$\psi^{*2}_{AB} = (\psi_A - \psi_B)^2$$

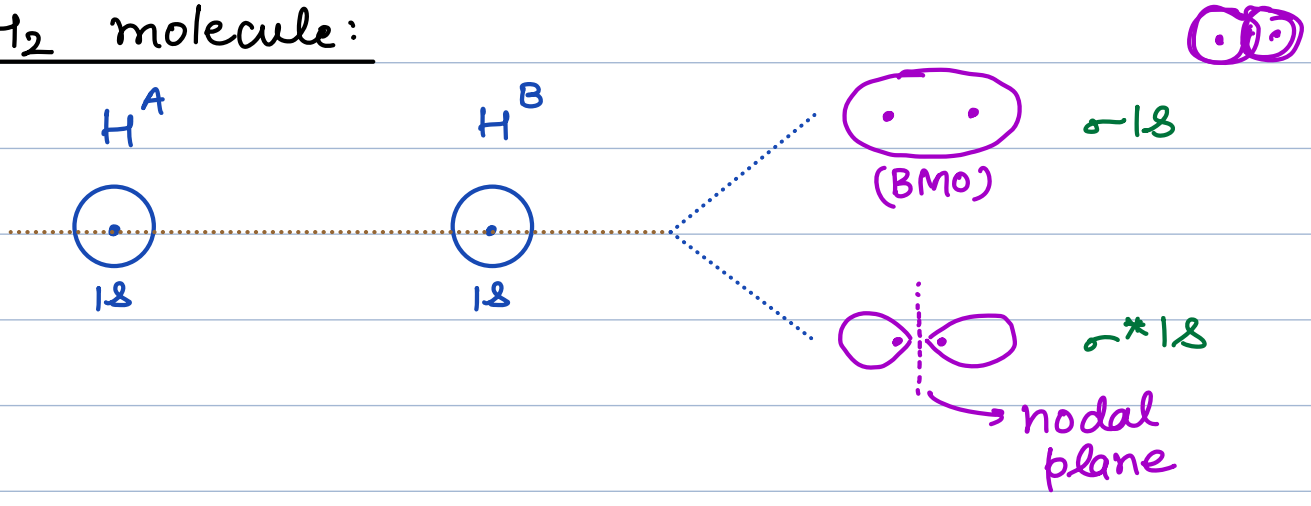
$$\psi^{*2}_{AB} = \psi_A^2 + \psi_B^2 - 2\psi_A\psi_B$$

(vii) The energy of bonding molecular orbitals (BMO) are lower than their respective parent atomic orbitals. The energy of anti-bonding molecular orbitals (ABMO) are higher than their respective parent atomic orbitals.

ABMO > Parent Atomic orbital > BMO

→ Energy

H_2 molecule:



In molecular orbitals:

(i) e⁻s are filled in inc. order of energy of MO's.

(ii) filling of e⁻ follows Hund's rule.

(iii) filling of e⁻ follows Pauli's principle.

Linear combination of Atomic orbitals (LCAO) to form molecular orbitals:

Conditions:

(a). Atomic orbitals must have equal or nearly equal energy.

(b). Atomic orbitals must have similar symmetry.

p_z —(can combine with)→ s (or) p_z (or) d_{z^2}

p_z —(can't combine with)→ p_x or p_y

(c). There should be the maximum extent of overlap.

Internuclear axis : Z-axis (INA)

Bond order (B.O.):

- No. of bonds b/w two atoms

- $B.O. = \frac{1}{2}\left(\text{no. of } e^- \text{s in BMO} - \text{no. of } e^- \text{s in ABMO}\right)$

- If, B.O. = Zero (or) -ve ⟶ species does not exist.

- If, B.O. = +ve ⟶ species stable and exists.

ex:
- $H_2(2)$: $\sigma 1s^2\ \sigma^* 1s^0$
  - B.O. $= \frac{1}{2}(2-0) = 1$
  - Diamagnetic

- $N_2(14) = \sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2$
  - B.O. $= \frac{1}{2}(10-4) = 3$
  - Diamagnetic

- $O_2(16)$:

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \pi^* 2p_x^2 = \pi^* 2p_y^0 < \sigma^* 2p_z$ → violation of Hund's

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \pi^* 2p_x^1 = \pi^* 2p_y^1 < \sigma^* 2p_z$

B.O. = 2

Paramagnetic (2 unpaired e^-)

- He_2 (4) : $\sigma 1s^2$ $\sigma^* 1s^2$

  $B.O. = \frac{1}{2}(2-2) = 0$

  i.e. He_2 does not exist.

- Be_2 (8) : $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$

  $B.O. = 0$

  i.e. Be_2 does not exist.

- N_2 (14): $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p_x^2$ $\sigma 2p_z^2$
  $\pi 2p_y^2$

  $B.O. = 3$ [σ - B.O. = 1, π - B.O. = 2]

- O_2 (16): $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\sigma 2p_z^2$ $\pi 2p_x^2$ $\pi^* 2p_x^1$
  $\pi 2p_y^2$ $\pi^* 2p_y^1$

  $B.O. = 2$ [σ - B.O. = 1, π - B.O. = 1]

- B_2 (10): $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p_x^1$
  $\pi 2p_y^1$

  $B.O. = 1$ (σ - B.O. = zero, π - B.O. = 1)

  Paramagnetic (2 unpaired e^-)

$C_2(12)$: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2$

B.O. $= 2$ (σ - B.O. $= 0$, π - B.O $= 2$)

Diamagnetic

Q. Compare bond length and bond dissociation energies in :

Dioxygen : O_2 | Peroxide ion : O_2^{2-}

superoxide ion : O_2^- | Dioxygenyl cation : O_2^+

Sol :

$O_2(16\ e^-)$: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^1 = \pi^* 2p_y^1$

B.O. $= 2$

$O_2^{2-}(18\ e^-)$: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^2 = \pi^* 2p_y^2$

B.O. $= 1$

$O_2^-(17\ e^-)$: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^2 = \pi^* 2p_y^1$

B.O. $= 1.5$

$O_2^+(15\ e^-)$: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^1 = \pi^* 2p_y$

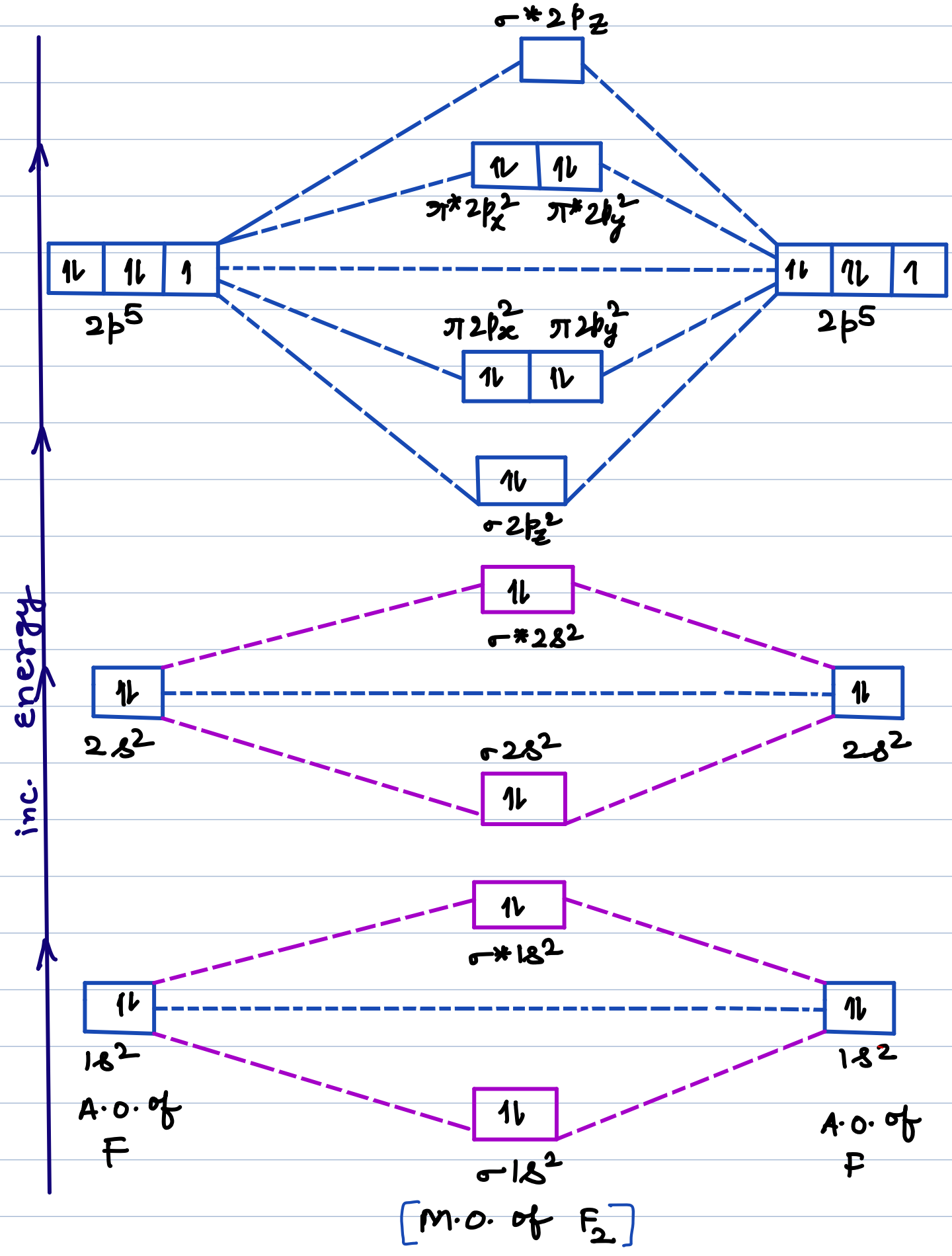
B.O. $= 2.5$

B.L. $\Rightarrow$ $O_2^{2-} > O_2^- > O_2 > O_2^+$

BDE $\Rightarrow$ $O_2^{2-} < O_2^- < O_2 < O_2^+$

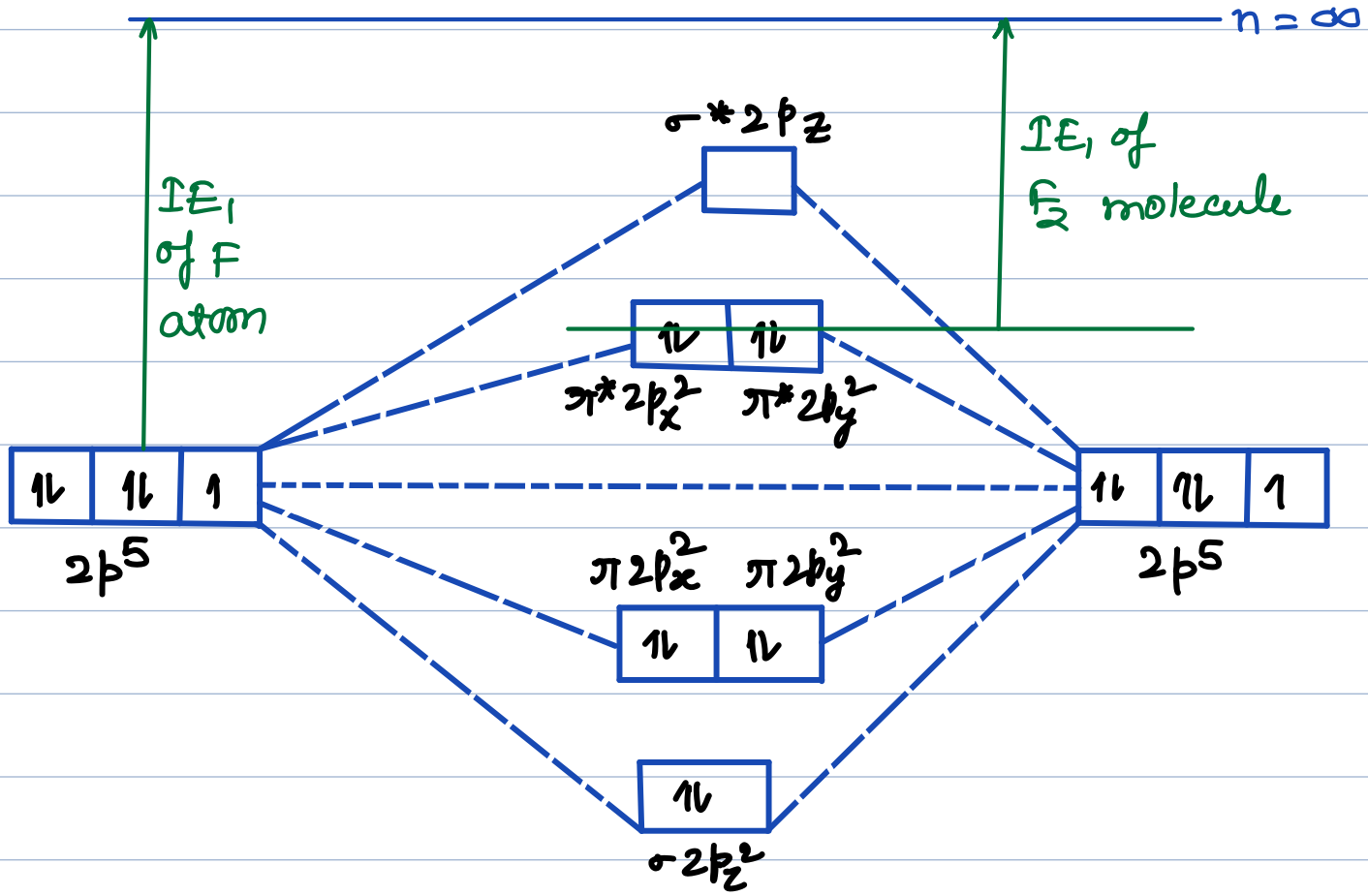
Molecular orbital Energy diagram :

ex: F_2



HOMO: Highest occupied Molecular orbital
LUMO: Lowest unoccupied Molecular orbital
SOMO: Singly occupied Molecular orbital

✓ For F_2 → HOMO = $\pi^* 2p_x$, $\pi^* 2p_y$
→ LUMO = $\sigma^* 2p_z$



✓ IE_1 : F atom > F_2 molecule

Results to be checked by yourself:

- IE_1 : H_2 > H
- IE_1 : O_2 < O
- IE_1 : N_2 > N

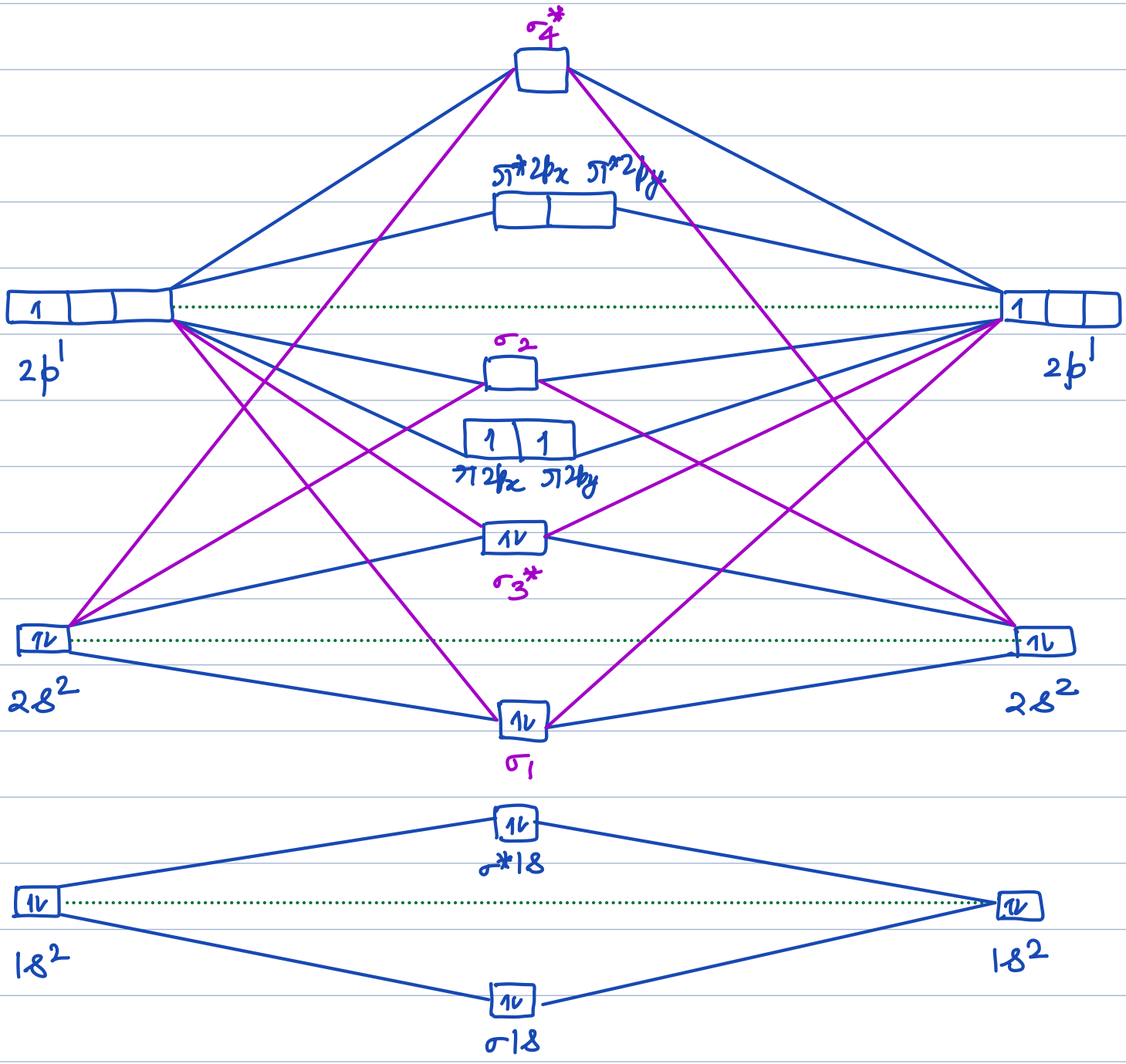
≠ Bond order can be -ve but only in excited state.

ex: $H_2(2)$: $\sigma 1s^0$ $\sigma^* 1s^2$

(excited)

$$B.O. = \frac{1}{2}(0-2) = -1$$

Exact molecular orbital energy diagram for B_2:



✓ usually, ionic bonds are formed between metallic and non-metallic elements.

✓ Electrovalency:

- The no. of e⁻s gained or lost in the formation of ionic bond is electrovalency.
- Charge is also considered in electrovalency.

ex: NaCl

electrovalency of Na = +1

electrovalency of Cl = −1

Species	electrovalency
Alkali metals	+1
Group-2 metals	+2
Halogens	−1
Chalcogens	−2

writing Formula of ionic compounds:

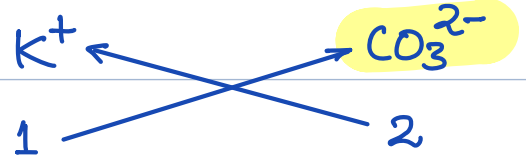
✓ Sodium nitrate

Na^+ ↖ ↗ NO_3^-

1 1

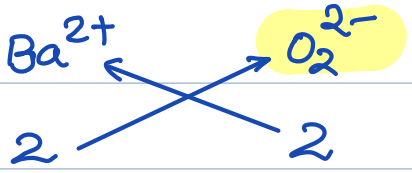
$\Rightarrow NaNO_3$

✓ Potassium carbonate



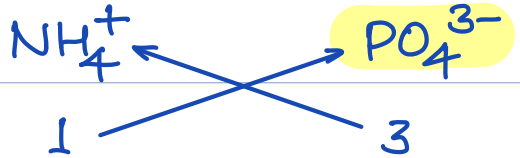
$\Rightarrow K_2CO_3$

✓ Barium peroxide



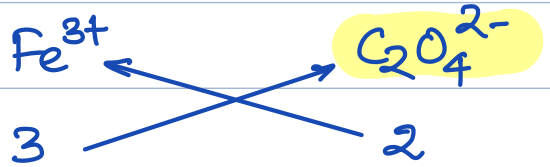
$\Rightarrow BaO_2$

✓ Ammonium phosphate



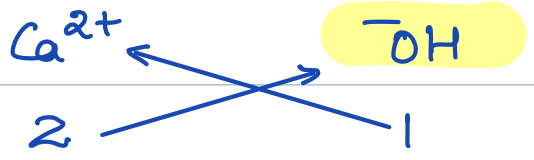
$\Rightarrow (NH_4)_3PO_4$

✓ Ferric oxalate



$\Rightarrow Fe_2(C_2O_4)_3$

✓ Calcium hydroxide



$\Rightarrow Ca(OH)_2$

Favourable conditions for ionic bond formation:

(i). High EN difference b/w elements

(ii). Low IE of element forming cation.

(iii). High EA of element forming anion

(iv). High Lattice energy of ionic compound

Lattice enthalpy: The enthalpy change on the formation of 1 mole of ionic solid from its constituent gaseous ions is called lattice enthalpy of that ionic solid.

$Na^+(g) + Cl^-(g) \longrightarrow NaCl(s)$; $\Delta H = -y$ KJ/mol

Lattice enthalpy of $NaCl(s) = -y$

(OR)

The enthalpy change on breaking of 1 mole ionic solid into constituent gaseous ions is called lattice enthalpy.

$NaCl(s) \longrightarrow Na^+(g) + Cl^-(g)$, $\Delta H = +y$ KJ/mol

Lattice enthalpy of $NaCl(s) = +y$

$\overset{+q_1}{A}\quad\overset{-q_2}{B}$: ionic compound

$$|\Delta_{lattice}H| \propto \frac{|q_1| \cdot |q_2|}{(r_{cation} + r_{anion})}$$

Compare $|\Delta_{lattice}H|$ in:

(i) $\overset{+1\ -1}{LiF} > \overset{+1\ -1}{NaF} > \overset{+1\ -1}{KF} > \overset{+1\ -1}{RbF} > \overset{+1\ -1}{CsF}$

(ii) $\overset{+2\ -2}{BeO} > \overset{+2\ -2}{MgO} > \overset{+2\ -2}{CaO} > \overset{+2\ -2}{SrO} > \overset{+2\ -2}{BaO}$

(iii) $\overset{+1\ -3}{Li_3N} < \overset{+2\ -3}{Mg_3N_2} < \overset{+3\ -3}{AlN}$

(iv) $\overset{+1\ -1}{NaF} < \overset{+2\ -2}{MgO}$ [dominance of charge factor]

(v) $\overset{+4\ -4}{TiC} > \overset{+3\ -3}{ScN} > \overset{+1\ 2-}{Li_2O}$ [dominance of charge factor]

(vi) $\overset{+1\ -1}{LiF} > \overset{+1\ -1}{LiCl} > \overset{+1\ -1}{LiBr} > \overset{+1\ -1}{LiI}$

Solid

Crystalline → Systematic arrangement of constituents

Amorphous → Irregular arrangement of constituents

Brittleness:

Polarisation in ionic bonds:

In ionic bond, cation attracts e^- cloud of anion towards itself. So the shape of anionic e^- cloud is distorted or deformed. It is termed as polarisation of anion by the cation.

"Due to polarisation, e^- prob. density inc. b/w nuclei of cation and anion so there is arise of covalent character in the ionic bond."

[inc. in e^- prob. density b/w two nuclei is property of covalent bond]

- Greater the extent of polarisation, more will be inc. of covalent nature in ionic bond.

Polarising power	Polarisability
✓ Power of cation to deform anion.	✓ Ability of anion to get distorted.
✓ Polarising power $\propto$ charge density of cation	Polarisability of anion $\propto$ |charge of anion| . (volume of anion)
✓ Polarising power $\propto \frac{\text{\|charge\|}}{\text{(volume of cation)}}$ → $\frac{4}{3}\pi r^3$	

Fajan's Rule: In an ionic bond, covalent character increases when:

(1). Cation is small.

(2). Anion is large.

(3). High charge on ions

(4). Cation has pseudo noble gas configuration.

pseudo noble gas configuration:

✓ Total 18 electrons in outermost shell

↳ $ns^2\ np^6\ nd^{10}$ configuration

Cation having such configuration → inc. Zeff due to poor shielding from d^{10} → inc. Polarising Power.

Cu: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

Cu^+: $1s^2\ 2s^2\ 2p^6$ $3s^2\ 3p^6\ 3d^{10}$ Pseudo noble gas config.

ex:

Cu^+	Zn^{+2}	Pb^{+4}	→ have pseudonoble gas config.
Ag^+	Cd^{+2}	Tl^{+3}	
Au^+	Hg^{+2}	Bi^{+5}	

Exp. Polarising power orders to remember:

(i): $Li^+ > Au^+ > Ag^+ > Cu^+ > Na^+ > K^+ > Rb^+ > Cs^+$

Zeff. basis

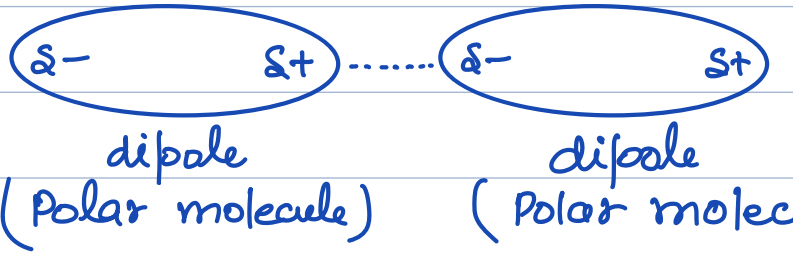
(ii): $Be^{+2} > Hg^{2+} > Cd^{2+} > Zn^{+2} > Mg^{+2} > Ca^{2+} > Sr^{2+} > Ba^{+2}$

Zeff basis

Weak Forces:

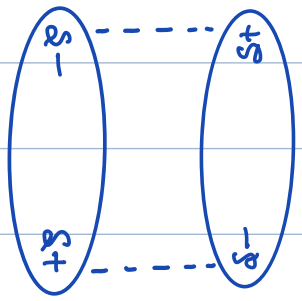
1. Dipole-dipole interaction [keesom forces]
2. Dipole - induced dipole interaction [Debye forces]
3. Instantantaneous dipole - induced dipole interaction
   [London dispersion forces ⇒ LDF]
   - → van der waal's force
   - → intermolecular forces
4. ion-dipole interaction
5. ion- induced dipole interaction

[1]. Dipole-dipole interaction:

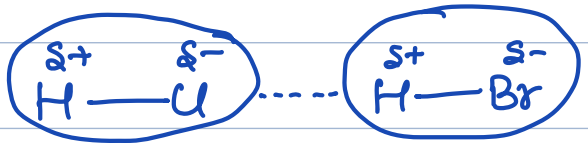


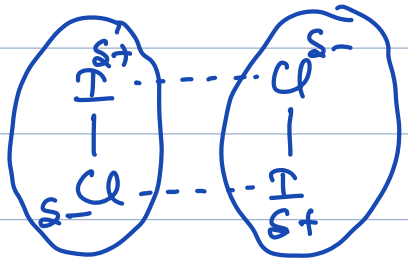
⇒ Head-tail arrangement

- when polar molecules come close together, their opposite charged terminals interact with each other through weak electrostatic force of attraction. It is known as dipole-dipole interaction.



→ Anti-parallel arrangement

ex: (i) 

(ii) 

- shorter the distance b/w molecules → stronger interaction.

- more the dipole moment of polar molecules, → more partial charges → stronger interaction

- | Interaction energy | → $1/d^3$ [stationary]
  → $1/d^6$ [rotating]

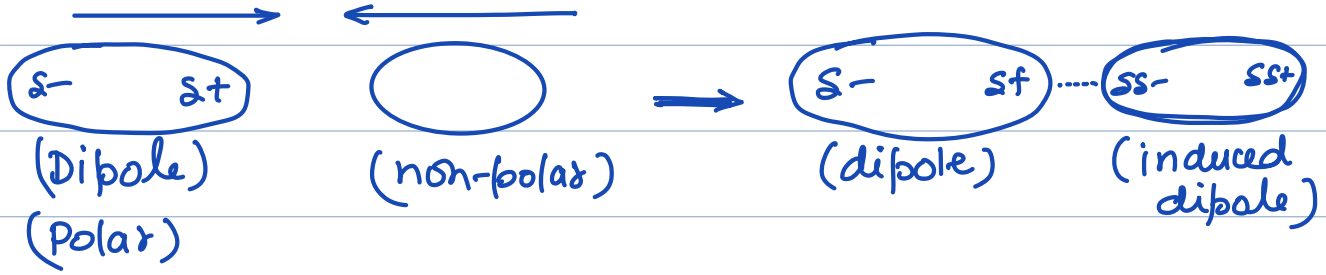
{d = distance b/w molecules}

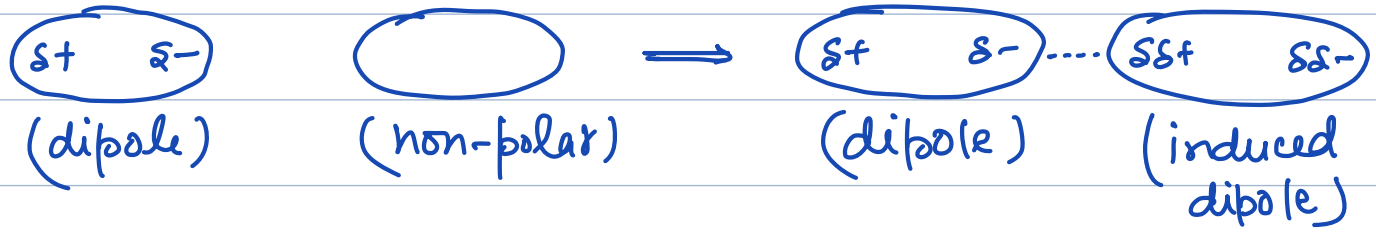
- solubility of polar covalent compounds in polar solvents is due to this interaction.

- H-bonding is a special case of this interaction.

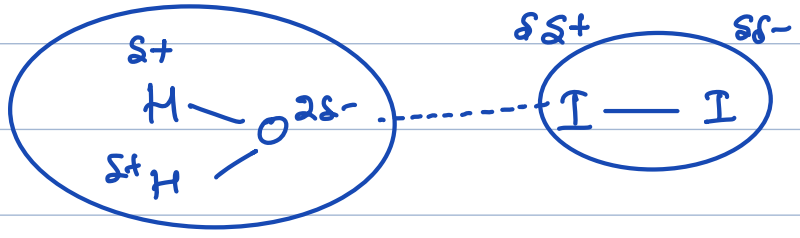
[2]. Dipole – induced dipole interaction:

when a non-polar molecule come closer to a polar molecule, charges are induced in non-polar (i.e. electron cloud of non-polar molecule is distorted). The weak interaction is developed between polar molecule and induced dipole, which is known as dipole-induced dipole interaction.





ex:



$$|\text{interaction energy}| \propto \frac{1}{d^6}$$

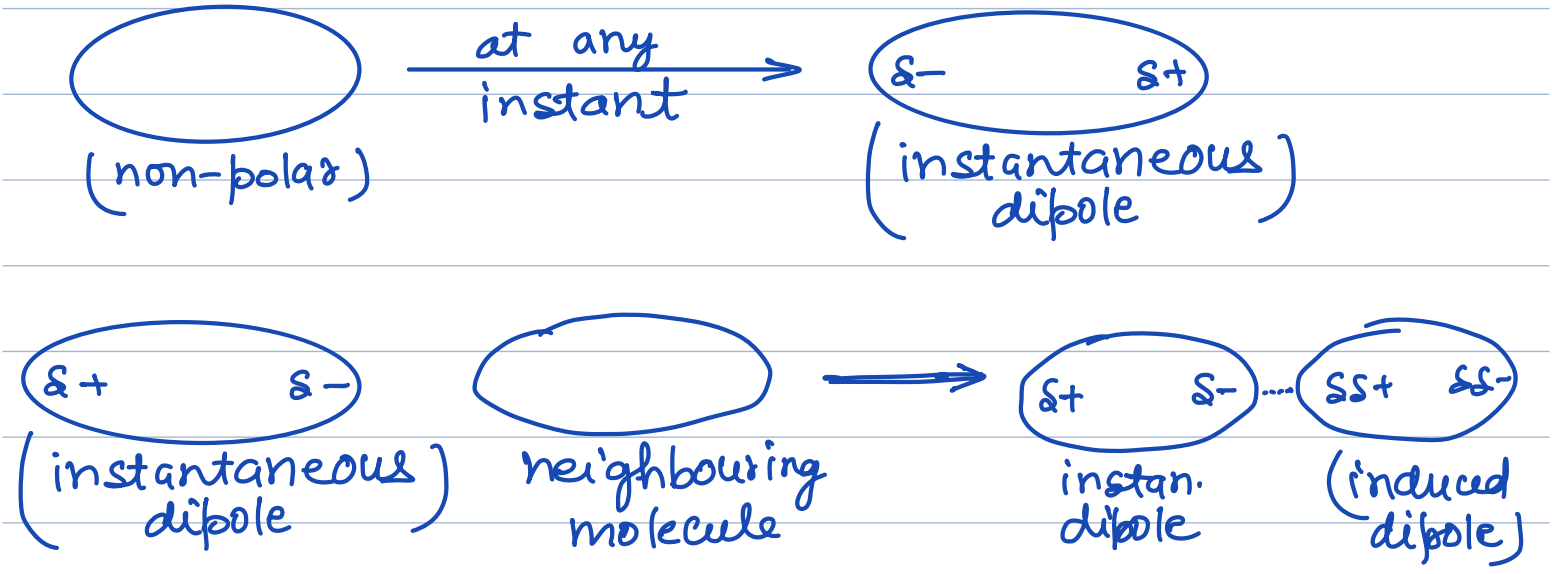
[d = distance b/w species]

[3] Instantaneous dipole - induced dipole interaction:

- In non-polar as well as polar molecules, electron cloud is not always symmetrically distributed within the molecule.

- At a particular instant, electron cloud is distorted, so that an instantaneous dipole is generated.

- This instantaneous dipole induces the charge in neighbouring molecules and a weak force of attraction is developed between them which is known as London dispersion forces.



**

LDF operates in all polar and non-polar molecules.

In b/w non-polar molecules, only LDF operates.

- $|\text{interaction energy}| \propto \frac{1}{d^6}$

****

no ionic bond is 100% ionic → due to polarisation.

no covalent bond is 100% covalent ⇒ due to EN difference or LDF

H---H $\xrightarrow[\text{data}]{\text{exp.}}$ 2% ionic character
98% covalent character

Factors affecting van der waal's force:

(i) size/polarisability:

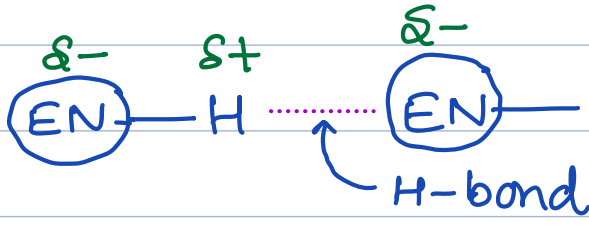
Larger size ⟶ more polarisability ⟶ more LDF

ex: $F_2(g)$ $Cl_2(g)$ $Br_2(l)$ $I_2(s)$
$O_2(g)$ $S_8(s)$
$N_2(g)$ $P_4(s)$

inc. size, inc. LDF →

Hydrogen Bond :

The attractive force which binds hydrogen atom which is already covalently bonded with highly electronegative atom to another electronegative atom is called the hydrogen bond (H-bond).



(EN = Generally F, O, N)

- H-bond is most simply regarded as a weak electrostatic interaction b/w a lone pair of electrons on highly electronegative atom and a covalently bonded H-atom which carries a partial charge δ+.

- In general, the range of dissociation energy of H-bond is 5 KJ/mol to 45 KJ/mol.

- Cl usually does not form H-bond due to its low charge density. Though there are examples where Cl forms intramolecular H-bonds with H attatched with highly EN atom.

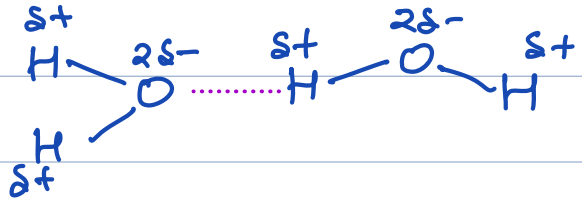
Types of H-bond:

[A]. Intermolecular H-bond:

H-bond b/w molecule to molecule

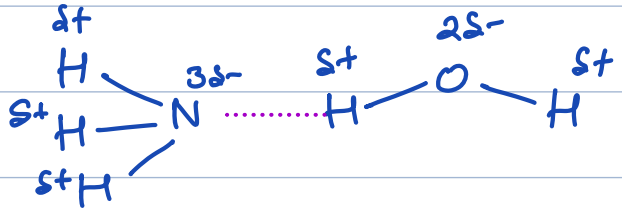
ex: (i) HF

(ii) H_2O

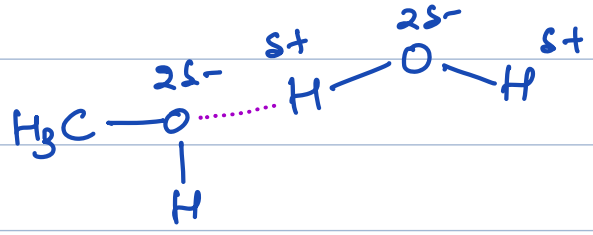


Homointermolecular H-bond

(iii) $NH_3 + H_2O$



(iv) $CH_3OH + H_2O$



Heterointermolecular H-bond

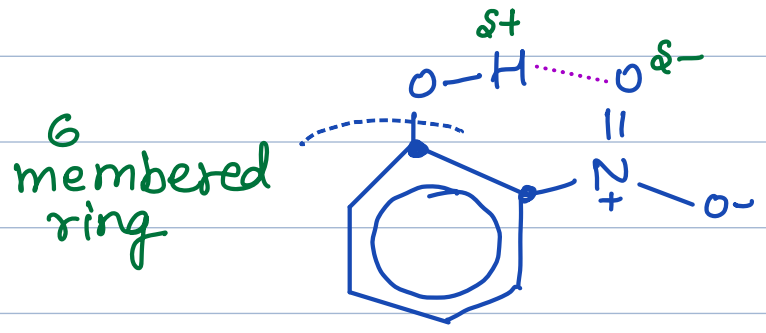
[B]. Intramolecular H-bond (chelation):

→ ring formation

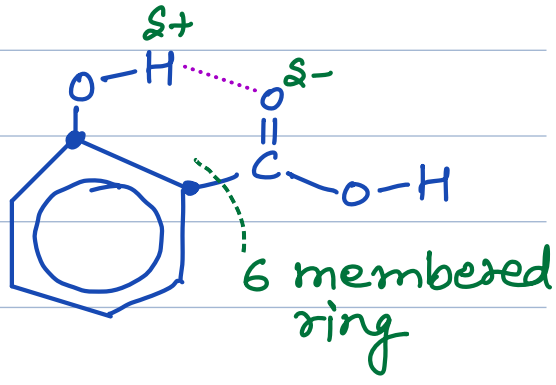
- H-bond formed within a molecule is intramolecular H-bond.

- usually 5 or 6 - membered rings are formed.

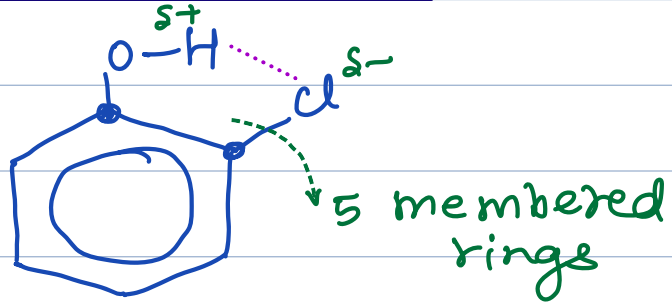
ex: (i) o-nitrophenol



(ii) o-hydroxybenzoic acid

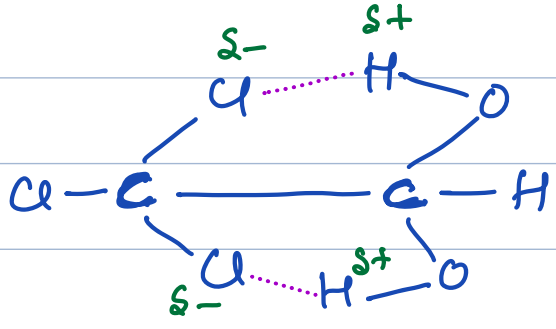


* (iii) o-chlorophenol:



(iv) chloral hydrate:

$(CCl_3)CH(OH)_2$



Consequences of intermolecular H-bonding:

(i). Generally increase of melting point and boiling point.

(ii) Low volatility

(iii). more viscosity

(iv). Low vapour pressure

ex: o-nitrophenol < p-nitrophenol (B.pt.)